

Measurement of bandstructure

8

8.1 Introduction

This chapter will describe a selection of the most common techniques used to measure the bandstructures of solids. Many of these techniques involve the use of magnetic fields to partially quantise the electronic motion; this is known as Landau quantisation or cyclotron motion. I shall therefore treat the motion of band electrons in some detail before giving brief details of the techniques themselves.

8.2 Lorentz force and orbits

8.2.1 General considerations

Equation 5.9 may be used to calculate the effect of an external magnetic field $\mathbf{B}$ on a band electron, i.e.

$$\hbar \frac{d\mathbf{k}}{dt} = -e\mathbf{v} \times \mathbf{B}, \quad (8.1)$$

where the right-hand side is the well-known Lorentz force. Equation 8.1 implies that

- the component of $\mathbf{k}$ parallel to $\mathbf{B}$ is constant;
- as $\frac{d\mathbf{k}}{dt}$ is perpendicular to $\mathbf{v}$ (defined property of the $\times$ operation) and as

$$\mathbf{v} = \frac{1}{\hbar} \nabla_{\mathbf{k}} E(\mathbf{k}) \quad (8.2)$$

(see eqn 5.5), $\frac{d\mathbf{k}}{dt}$ is perpendicular to $\nabla_{\mathbf{k}} E(\mathbf{k})$. This means that the electron path (orbit) is one of constant energy. To see this, consider $\frac{d\mathbf{k}}{dt} \cdot \nabla_{\mathbf{k}} E(\mathbf{k}) = 0$ (because the two are perpendicular); writing this in components gives $\frac{\partial k_x}{\partial t} \frac{\partial E}{\partial k_x} + \frac{\partial k_y}{\partial t} \frac{\partial E}{\partial k_y} + \frac{\partial k_z}{\partial t} \frac{\partial E}{\partial k_z} \equiv \frac{dE}{dt}$ from the chain rule. Hence, $\frac{dE}{dt} = 0$.

In k -space, the possible electron orbits are therefore described by the intersections of surfaces of constant energy with planes perpendicular to $\mathbf{B}$ (see also Exercise 8.2).

8.2.2 The cyclotron frequency

Consider a section of k -space orbit of constant energy E in a plane perpendicular to $\mathbf{B}$ (see Fig. 8.1). The time $t_2 - t_1$ taken to traverse the part of

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the orbit between $\mathbf{k}_1$ and $\mathbf{k}_2$ is

$$t_2 - t_1 = \int_{t_1}^{t_2} dt = \int_{\mathbf{k}_1}^{\mathbf{k}_2} \frac{dk}{|\dot{\mathbf{k}}|}, \quad (8.3)$$

where $\dot{\mathbf{k}} = \frac{d\mathbf{k}}{dt}$. Equations 8.1 and 8.2 can be used to obtain $\dot{\mathbf{k}}$, so that eqn 8.3 becomes

$$t_2 - t_1 = \frac{\hbar^2}{eB} \int_{\mathbf{k}_1}^{\mathbf{k}_2} \frac{dk}{|\nabla_{\mathbf{k}\perp} E|}, \quad (8.4)$$

where $\nabla_{\mathbf{k}\perp} E$ is the component of $\nabla_{\mathbf{k}} E$ perpendicular to the field.

The quantity $\nabla_{\mathbf{k}\perp} E$ might seem rather nebulous;¹ however, it has a simple geometrical interpretation. Consider a second orbit in the same plane as the one of energy E defined at the start of this section, but with energy $E + \delta E$ (see Fig. 8.1). Let the vector $\delta\mathbf{k}$ join a point $\mathbf{k}$ on the orbit of energy E to a point on the orbit of energy $E + \delta E$, and let it also be perpendicular to the orbit of energy E at $\mathbf{k}$. If δE is small, then

$$\delta E = \nabla_{\mathbf{k}} E \cdot \delta\mathbf{k} = \nabla_{\mathbf{k}\perp} E \cdot \delta\mathbf{k}. \quad (8.5)$$

As $\nabla_{\mathbf{k}}$ is perpendicular to surfaces of constant energy, then $\nabla_{\mathbf{k}\perp}$ is perpendicular to the orbit of energy E and therefore parallel to $\delta\mathbf{k}$. Equation 8.5 becomes

$$\delta E = |\nabla_{\mathbf{k}\perp} E| \delta k, \quad (8.6)$$

so that eqn 8.4 can be rewritten

$$t_2 - t_1 = \frac{\hbar^2}{eB} \frac{1}{\delta E} \int_{\mathbf{k}_1}^{\mathbf{k}_2} \delta k dk. \quad (8.7)$$

The integral in eqn 8.7 is the area of the k -space plane between orbits E and $E + \delta E$; therefore if $\delta E \rightarrow 0$ then

$$t_2 - t_1 = \frac{\hbar^2}{eB} \frac{\partial A_{1,2}}{\partial E}, \quad (8.8)$$

where $\frac{\partial A_{1,2}}{\partial E}$ is the rate at which the orbit from $\mathbf{k}_1$ to $\mathbf{k}_2$ sweeps out area in k -space as E increases.

In most cases it is going to be useful to work in terms of closed orbits (cyclotron orbits) in k -space, i.e. closed paths where $\mathbf{k}_1 = \mathbf{k}_2$. In this case A becomes the area in k -space of the closed orbit; this will depend on E and the component of $\mathbf{k}$ parallel to $\mathbf{B}$, which we denote $k_{\parallel}$. The period τ_c of the closed orbit is

$$\tau_c = \frac{\hbar^2}{eB} \frac{\partial A(E, k_{\parallel})}{\partial E}; \quad (8.9)$$

turning this into an angular frequency ω_c (the cyclotron frequency), we obtain

$$\omega_c = \frac{eB}{m_{\text{CR}}^*}, \quad (8.10)$$

where

$$m_{\text{CR}}^* = \frac{\hbar^2}{2\pi} \frac{\partial A(E, k_{\parallel})}{\partial E}. \quad (8.11)$$

The quantity m_{CR}^* defined in eqn 8.11 is known as the cyclotron mass. It is easy to show that for a free-electron system this quantity is just m_e , so that eqn 8.10 yields the classical cyclotron frequency $\omega_c = eB/m_e$.² Similarly, in the case of a constant, isotropic effective mass m^* , it can be shown in a straightforward manner that $\omega_c = eB/m^*$.

¹ Or even 'nebulous'.

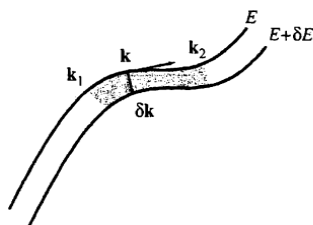


Fig. 8.1 Geometrical interpretation of parameters used in the derivation of the cyclotron frequency; the two curves represent constant energy orbits of energy E and $E + \delta E$ in the plane perpendicular to the magnetic field. The electron traverses from $\mathbf{k}_1$ to $\mathbf{k}_2$ on the orbit of energy E ; $\delta\mathbf{k}$ is perpendicular to the orbit of energy E and connects it to the orbit of energy $E + \delta E$. The small arrow perpendicular to $\delta\mathbf{k}$ shows the instantaneous direction of $d\mathbf{k}$ which runs along the orbit in the integral of eqn 8.4. The shaded area is the area of the k -space plane between orbits E and $E + \delta E$.

² See, for example, *Electricity and magnetism*, by B.I. Bleaney and B. Bleaney, third edition (Oxford University Press, Oxford, 1989) page 126.

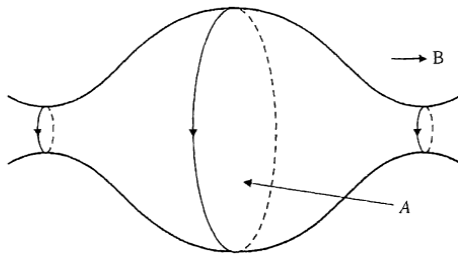


Fig. 8.2 Orbits on a Fermi surface section are in planes perpendicular to **B**. Here, *A* represents the *k*-space cross-sectional area of the orbit.

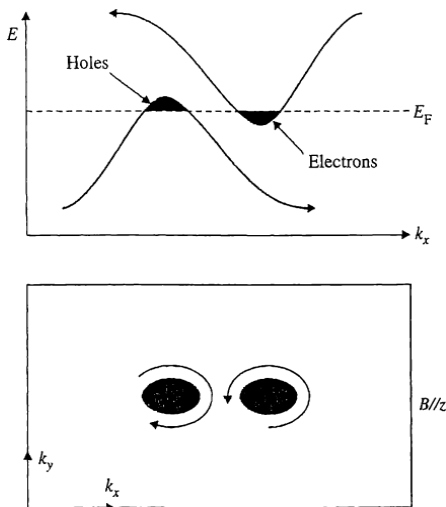


Fig. 8.3 The use of the sign of $(\partial A/\partial E)$ to identify a section of Fermi surface as hole-like or electron-like. The top half of the figure shows a schematic bandstructure with two bands crossing the Fermi energy E_F . E_F crosses the left-hand band near its maximum; the band is almost full, with holes (empty states) at the top of the band surrounded by filled states. E_F crosses the right-hand band near its minimum, so that there are electrons in the bottom of the band surrounded by empty states. The lower half of the figure shows the corresponding Fermi-surface cross-sections. In the left-hand case, as E_F increases, the area of the section of Fermi surface decreases (hole-like). In the right-hand case, as E_F increases, the area of the section of Fermi surface increases (electron-like). The arrows show the opposite senses of cyclotron motion about the Fermi-surface sections.

8.2.3 Orbits on a Fermi surface

Chapters 1–5 showed that the Fermi surface of a metal is a constant-energy surface *par excellence*. Therefore, if the material subjected to the magnetic field is metallic (i.e. possesses sections of Fermi surface), then Section 8.2.1 implies that the electrons will perform orbits in *k*-space about Fermi-surface cross-sections perpendicular to **B** (see Fig. 8.2). Equations 8.10 and 8.11 can therefore give information about the cross-sectional areas of a section of Fermi surface. Note that the *sign* of $(\partial A/\partial E)$ allows the carriers on a section of Fermi surface to be identified as hole-like ($(\partial A/\partial E) < 0$) or electron-like ($(\partial A/\partial E) > 0$) in a very appealing manner (see Fig. 8.3).

8.3 The introduction of quantum mechanics

8.3.1 Landau levels

A general proof of the quantised motion of a carrier in a magnetic field in an arbitrarily-shaped band is difficult. In this section, I shall use a band defined by the effective-mass tensor of Exercise 5.2, Chapter 5, with its minimum centred on $k = 0$ (a good approximation to many of the band extrema in

semiconductors) and derive some analytical solutions for the eigenenergies of the electrons (the Landau levels). The anisotropic band also yields some useful insights about generalised bandstructure.

For further generality, Section 8.3.2 will derive the Landau quantisation of arbitrarily-shaped bands in the limit that $\hbar\omega_c \ll E_F$.

The Hamiltonian for electrons in the band defined by the effective mass tensor of Exercise 5.2, with its minimum centred on $k = 0$ is

$$\left\{ \frac{\mathcal{P}_x \mathbf{e}_1}{(2m_1)^{\frac{1}{2}}} + \frac{\mathcal{P}_y \mathbf{e}_2}{(2m_2)^{\frac{1}{2}}} + \frac{\mathcal{P}_z \mathbf{e}_3}{(2m_3)^{\frac{1}{2}}} \right\}^2 \psi = E\psi, \quad (8.12)$$

where the $\mathcal{P}$ are one-dimensional momentum operators; i.e. all of the effects of the crystalline potential have been absorbed into the bandstructure, which is defined by the effective masses. Substitution of a plane wave solution into eqn 8.12 yields energies of the correct form, i.e.

$$E = \frac{\hbar^2 k_x^2}{2m_1} + \frac{\hbar^2 k_y^2}{2m_2} + \frac{\hbar^2 k_z^2}{2m_3}. \quad (8.13)$$

Consider a magnetic field $\mathbf{B}$ directed along the z ($\mathbf{e}_3$) axis. In addition, remember that $\mathbf{B} = \nabla \times \mathbf{A}$, where $\mathbf{A}$ is the magnetic vector potential. One particular $\mathbf{A}$ which gives the correct $\mathbf{B}$ is

$$\mathbf{A} = (0, Bx, 0); \quad (8.14)$$

this choice is known as the Landau gauge. For an electron in a magnetic field,

$$\mathcal{P} \rightarrow (\mathcal{P} + e\mathbf{A}) \quad (8.15)$$

(see Appendix G). Making such a substitution into eqn 8.12 gives

$$\left\{ \frac{\mathcal{P}_x \mathbf{e}_1}{(2m_1)^{\frac{1}{2}}} + \frac{(\mathcal{P}_y + eBx) \mathbf{e}_2}{(2m_2)^{\frac{1}{2}}} + \frac{\mathcal{P}_z \mathbf{e}_3}{(2m_3)^{\frac{1}{2}}} \right\}^2 \psi = E\psi. \quad (8.16)$$

All that we have done is introduce an extra term with x in it; as $[\mathcal{P}_y, x] = [\mathcal{P}_z, x] = 0$,³ $\mathcal{P}_y$ and $\mathcal{P}_z$ still commute with the Hamiltonian; their associated physical quantities are thus constants of the motion. The operators $\mathcal{P}_y$ and $\mathcal{P}_z$ in eqn 8.16 can therefore be replaced by their constant values $\hbar k_y$ and $\hbar k_z$ respectively. Equation 8.16 becomes

$$\left\{ \frac{\mathcal{P}_x \mathbf{e}_1}{(2m_1)^{\frac{1}{2}}} + \frac{(\hbar k_y + eBx) \mathbf{e}_2}{(2m_2)^{\frac{1}{2}}} + \frac{\hbar k_z \mathbf{e}_3}{(2m_3)^{\frac{1}{2}}} \right\}^2 \psi = E\psi. \quad (8.17)$$

Squaring the bracket, remembering that the Cartesian unit vectors $\mathbf{e}_j$ are mutually perpendicular (i.e. $\mathbf{e}_j \cdot \mathbf{e}_l = \delta_{jl}$), gives

$$\left\{ \frac{\mathcal{P}_x^2}{2m_1} + \frac{(\hbar k_y + eBx)^2}{2m_2} + \frac{\hbar^2 k_z^2}{2m_3} \right\} \psi = E\psi. \quad (8.18)$$

Making the substitutions $E' = E - (\hbar^2 k_z^2 / 2m_3)$ and $x_0 = -(\hbar k_y / eB)$, and rearranging yields

$$\left\{ \frac{\mathcal{P}_x^2}{2m_1} + \frac{e^2 B^2}{2m_2} (x - x_0)^2 \right\} \psi = E' \psi. \quad (8.19)$$

³These are standard commutation relationships; see e.g. *Quantum mechanics*, by Stephen Gasiorowicz (Wiley, New York, 1974) pages 51, 141.

Equation 8.19 looks exactly like the Hamiltonian of a one-dimensional harmonic oscillator, i.e.,

$$\left\{ \frac{\mathcal{P}_x^2}{2m} + \frac{1}{2}m\omega^2(x - x_0)^2 \right\} \psi = E' \psi, \quad (8.20)$$

with $\omega = eB/(m_1m_2)^{\frac{1}{2}}$; thus $E' = (l + \frac{1}{2})\hbar\omega$, with $l = 0, 1, 2, 3, \dots$. The energy levels of the electron are therefore

$$E(l, B, k_z) = \frac{\hbar^2 k_z^2}{2m_3} + (l + \frac{1}{2})\hbar\omega, \quad (8.21)$$

with

$$\omega = eB/(m_1m_2)^{\frac{1}{2}} \equiv \omega_c. \quad (8.22)$$

The equivalence of ω and ω_c , the cyclotron frequency for this geometry (i.e.

$\omega_c = eB/m_{\text{CR}}^*$, with $m_{\text{CR}}^* = \frac{\hbar^2}{2\pi} \frac{\partial A(E, k_{\parallel})}{\partial E}$), is proved in Exercise 8.3.

The following points may be deduced from eqn 8.21.

- The energy of the electron's motion in the plane perpendicular to **B** is completely quantised. These quantised levels are known as *Landau levels*.
- The k -space areas of the orbits in the plane perpendicular to **B** are also quantised (this is easy to work out in the present case and is left as an exercise); thus, allowed orbits fall on 'Landau tubes' in k -space with quantised cross-sectional area.
- The energy 'quantum' for the in-plane motion appears to be $\hbar \times$ (the semiclassical cyclotron frequency).
- The motion parallel to **B** is unaffected (cf. Section 8.2.1).

We shall consider the second point in more detail (and more generally) in the following section.

8.3.2 Application of Bohr's correspondence principle to arbitrarily-shaped Fermi surfaces in a magnetic field

In Section 8.4 below we are going to consider the effect of the quantisation of the k -space orbits caused by **B** on the Fermi surfaces of metals; i.e. we shall be dealing with Landau levels which cut the Fermi surface. The Fermi surfaces of real metals are in general not as simple as the case dealt with in the previous section. Therefore, in order to treat an arbitrarily-shaped band, resulting in an arbitrarily-shaped Fermi surface, we shall use Bohr's correspondence principle, which states in this context that the difference in energy of two adjacent levels is $\hbar$ times the angular frequency of classical motion at the energy of the levels.

The correspondence principle is only valid for levels with very large quantum numbers. However, this is not really a problem, as we are going to be dealing with Landau levels with energies comparable to E_F ; in most metals E_F is $\sim$ several eV, whereas the cyclotron energy of free electrons is $\hbar eB/m_e \approx 1.16 \times 10^{-4}$ eV at $B = 1$ T. Laboratory fields are usually ~ 10 T,

and so the quantum numbers of Landau levels with energies $\sim E_F$ will be $l \sim 10^4$; the correspondence principle should work reasonably well.

The classical frequency at the energy of interest (i.e. E_F) is given by eqn 8.10

$$\omega_c = \frac{eB}{m_{CR}^*}, \quad (8.23)$$

with m_{CR}^* given by eqn 8.11 as follows

$$m_{CR}^* = \frac{\hbar^2}{2\pi} \frac{\partial A(E, k_{||})}{\partial E}. \quad (8.24)$$

Here, A is now specifically *the cross-sectional area of the Fermi surface in a plane perpendicular to $\mathbf{B}$* . Thus, the separation in energy of two Landau tubes with energies close to E_F is

$$E(l+1, B, k_{||}) - E(l, B, k_{||}) = \frac{\hbar e B}{m_{CR}^*}, \quad (8.25)$$

with m_{CR}^* defined by eqn 8.24. (It is interesting to compare this equation with eqn 8.21.)

8.3.3 Quantisation of the orbit area

Equation 8.25 may be rewritten as

$$(E(l+1, B, k_{||}) - E(l, B, k_{||})) \frac{\partial A(E)}{\partial E} = \frac{2\pi e B}{\hbar}. \quad (8.26)$$

Now as the E in $A(E)$ is of order E_F , then the difference between adjacent levels will be $\ll$ the energies of the levels themselves. Thus the approximation

$$\frac{\partial A(E)}{\partial E} = \frac{A(E(l+1, B, k_{||})) - A(E(l, B, k_{||}))}{E(l+1, B, k_{||}) - E(l, B, k_{||})} \quad (8.27)$$

may be made, so that eqn 8.26 becomes

$$A(E(l+1, B, k_{||})) - A(E(l, B, k_{||})) = \frac{2\pi e B}{\hbar}. \quad (8.28)$$

This states that classical orbits at adjacent allowed energies and the same $k_{||}$ enclose k -space areas that differ by a fixed amount δA , with

$$\delta A = \frac{2\pi e B}{\hbar}. \quad (8.29)$$

Putting this another way, the k -space area enclosed by an orbit of allowed energy and $k_{||}$ must be given by

$$A(E(l+1, B, k_{||})) = (l + \lambda)\delta A = (l + \lambda) \frac{2\pi e B}{\hbar} \quad (8.30)$$

for large l , where λ is a constant ~ 1 .

As in Section 8.3.1, we have found 'Landau tubes' which define the allowed k -space orbits; rather than being quasicontinuous, the Fermi surface will now be composed of occupied states lying on Landau tubes (see Fig. 8.4). Note that the k -space area of each tube increases with B ; as we shall see in the sections below, this means that tubes 'pop out' of the Fermi surface as B increases.

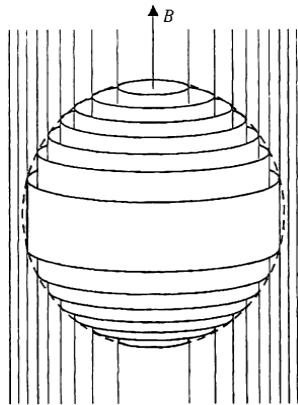


Fig. 8.4 Schematic of a Fermi sphere rearranged into Landau tubes.

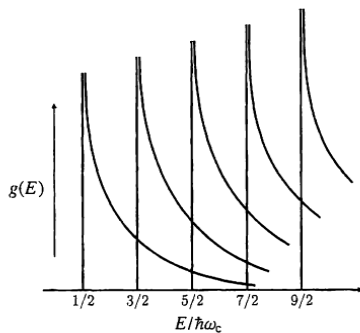


Fig. 8.5 Schematic of the electronic density of states in a magnetic field. The 'sawtooth' structure results from a one-dimensional density of states (from the motion parallel to $\mathbf{B}$) superimposed on the δ -function-like Landau levels.

8.3.4 The electronic density of states in a magnetic field

The above sections have shown that the energy of the band electron is completely quantised into Landau levels in the plane perpendicular to $\mathbf{B}$, whilst the motion parallel to $\mathbf{B}$ remains unconstrained. This means that the density of states of the electrons is an infinite 'ladder' of Landau levels, each with a one-dimensional density of states function, due to the motion parallel to $\mathbf{B}$, superimposed.⁴ The overall density of states will therefore be of the form $g(E) = \sum_{l=0}^{\infty} C(E - E(l, B))^{-1/2}$, where $E(l, B)$ is the energy of the Landau level (i.e. the lowest energy point on the Landau tube) and C is a constant; it is shown schematically in Fig. 8.5.

⁴In other words, as far as the density of states is concerned, the magnetic field reduces the effective dimensionality of the electron system by 2.

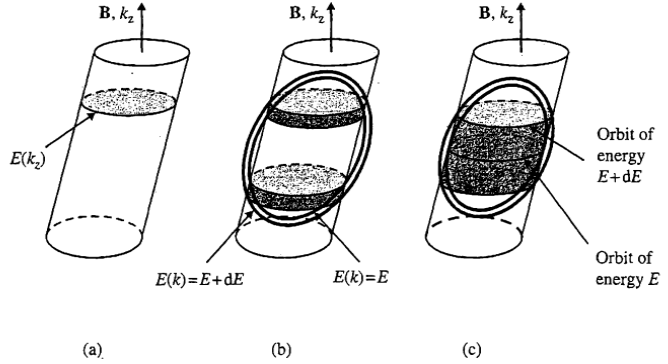
8.4 Quantum oscillatory phenomena

Each time one of the sharp peaks in the electronic density of states moves through the chemical potential μ ,⁵ there will be a modulation of the density of states at μ . Since almost all of a metal's properties depend on the density of states at μ , we expect this to affect the behaviour of the metal in some way.

Figure 8.6 shows a Landau tube superimposed on constant energy surfaces E and $E + \delta E$. The left-hand part of the figure shows the general case; as

⁵Remember from earlier chapters that $E_F \equiv \mu(T = 0)$. In many books, quantum oscillations are said to result from 'Landau levels passing through the Fermi energy'. This is only strictly true at $T = 0$; the chemical potential is the important energy as far as the electrons are concerned. In metals at low temperatures there will be a very small difference between E_F and μ ; nevertheless, one should not tolerate sloppy terminology.

Fig. 8.6 Schematic of Landau tubes crossing surfaces of constant energy E and $E + \delta E$. (a) shows the Landau tube on its own; (b) shows the Landau tube intersecting constant energy surfaces E and $E + \delta E$ when the tube does not correspond to an extremal orbit; (c) shows the Landau tube intersecting constant energy surfaces E and $E + \delta E$ when the tube corresponds to an extremal orbit. In (c), a much greater area of the cylinder (and therefore a larger number of states) lies between E and $E + \delta E$ than is the case in (b).



the field increases, the tube sweeps outwards, so that some of its states are always crossing the constant energy surfaces. However, in the right-hand part of the figure, the tube reaches an **extremal** cross-section of the constant energy surface E . In this case, a much larger area of the cylinder (and therefore a much greater number of states, which lie uniformly in k -space) lies between E and $E + \delta E$. Returning to the case of the Fermi surface, we can therefore say that the maximum effect of the Landau quantisation will occur when a Landau tube (and its associated peak in the density of states) crosses an extremal cross-section of a Fermi surface. By analogy with the discussion of classical orbits about the Fermi surface in Section 8.2.1, such cross-sections are often known as **extremal orbits**.

The Landau tube crosses an extremal orbit/cross-section when

$$(l + \lambda) \frac{2\pi e B}{\hbar} = A_{\text{ext}}, \quad (8.31)$$

where A_{ext} is the k -space area of the extremal cross-section of the Fermi surface in a plane perpendicular to $\mathbf{B}$. Thus, the metal's properties will oscillate as $\mathbf{B}$ changes, with a period given by

$$\Delta \left(\frac{1}{B} \right) = \frac{2\pi e}{\hbar} \frac{1}{A_{\text{ext}}}. \quad (8.32)$$

Such oscillations are known generically as **magnetic quantum oscillations**.

Often it is more convenient to describe a series of quantum oscillations using a frequency F , also known as the **fundamental field**, B_F ;

$$F \equiv B_F = \frac{1}{\Delta(1/B)} = \frac{\hbar}{2\pi e} A_{\text{ext}}. \quad (8.33)$$

Note that

- the fundamental field of the quantum oscillations is determined solely by the Fermi surface extremal area and fundamental constants;
- several different frequencies (fundamental fields) of quantum oscillation may be simultaneously present for a given orientation of $\mathbf{B}$, corresponding to different possible extremal orbits (see Fig. 8.7); a famous example is the simultaneous observation of frequencies from 'neck and belly' orbits in Cu and Au;

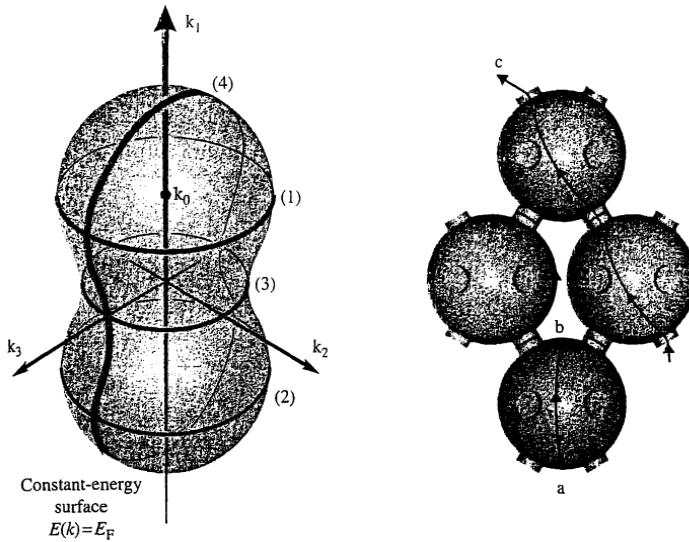


Fig. 8.7 Illustration of the types of extremal orbit about Fermi surfaces which can give rise to quantum oscillations. The left-hand figure shows that for $\mathbf{B}$ parallel to k_1 there will be three extremal orbits ((1), (2) and (3)); as (1) and (2) have the same area, there will be two series of quantum oscillation (i.e. two different frequencies will be present). However, for $\mathbf{B}$ parallel to k_2 , there is only one extremal orbit (4); a similar situation would apply if $\mathbf{B}$ were parallel to k_3 . The right-hand figure helps to visualise the different types of orbit possible in metals possessing necks and bellies in their Fermi surfaces, such as copper, silver or gold. The orbit labelled a is an orbit around the 'belly' of the Fermi surface. If the magnetic field is applied out of the page, then an extremal 'hole-like' orbit (labelled b) will occur around the bone-shaped space in the middle; this is known as the dog's bone orbit! In orbit c, electrons run across the bellies and necks in an extended or open orbit; *this does not give rise to quantum oscillations*, as there is no closed orbit with which a frequency can be associated. If the magnetic field is applied along one of the neck directions, extremal orbits about both neck and belly are possible, leading to two frequencies of quantum oscillations. Note that different field orientations are required to observe the different cases; the orbits in k -space are *always* in planes perpendicular to $\mathbf{B}$.

- a measurement of the observed frequencies as a function of magnetic field orientation allows the Fermi-surface shape to be mapped out;
- open orbits do not give rise to quantum oscillations.⁶

8.4.1 Types of quantum oscillation

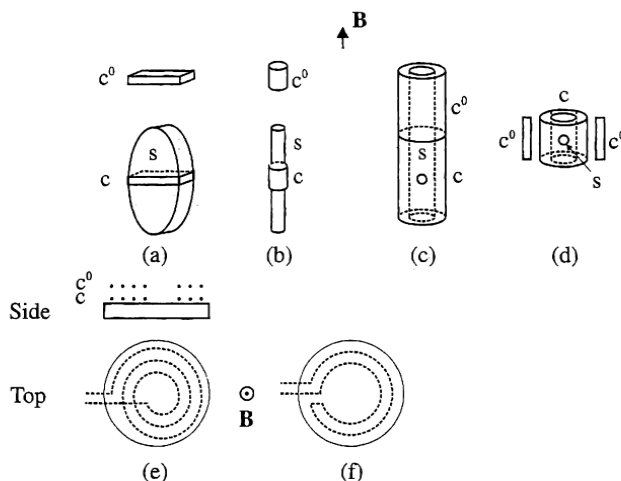
As the electronic density of states at E_F determines most of a metal's properties, virtually all properties will exhibit quantum oscillations in a magnetic field. Examples include⁷

- oscillations of the magnetisation (the de Haas–van Alphen effect);
- oscillations of the magnetoresistance (the Shubnikov–de Haas effect);
- oscillations of the sample length;
- oscillations of the sample temperature;

⁶However, open orbits do lead to a very interesting quantum phenomenon which has recently been observed in high-frequency experiments; see A. Ardavan *et al.*, *Phys. Rev. B* **60**, 15500 (1999); *Phys. Rev. Lett.* **81**, 713 (1998).

⁷Some pictures of typical data are shown in *Solid state physics*, by N.W. Ashcroft and N.D. Mermin (Holt, Rinehart and Winston, New York, 1976) pages 266–268.

Fig. 8.8 Coils for the observation of the de Haas–van Alphen effect. For each set, c is the coil flux-linked to the sample and c^0 is the compensating coil, only weakly flux-linked to the sample: (a) c is wound around the centre of a disc-like sample; (b) c is wound around a sample cut into a cylinder; (c) a small sample inside colinear coils c and c^0 ; (d) small sample inside coaxial coils c and c^0 ; (e) side and top views of a coil evaporated down onto a thin sample; (f) as (e) but with only a single annular turn. In both (e) and (f) c^0 has not been shown. (Based on figures in *Magnetic oscillations in metals*, by David Shoenberg (Cambridge University Press, Cambridge, 1984).)



- oscillations in the ultrasonic attenuation;
- oscillations in the Peltier effect and thermoelectric voltage;
- oscillations in the thermal conductivity.

8.4.2 The de Haas–van Alphen effect

The de Haas–van Alphen effect is perhaps the most significant of the quantum oscillatory phenomena in metals; as the magnetisation M is a thermodynamic function of state,⁸ it can be directly related to the density of states and Fermi–Dirac distribution function of the electrons in a metal, *without additional assumptions*. This means that theoretical models for the Fermi surface can be checked in a very rigorous manner.⁹

Figure 8.8 shows a schematic of the types of coil most commonly used to observe the de Haas–van Alphen effect. Each system consists of a coil c which contains the sample and a *compensating coil* c^0 ; both are placed in a time-varying magnetic field. The coils c and c^0 are connected in series-opposition and carefully balanced so that in the absence of a sample the voltage induced by dB/dt in c^0 exactly cancels that in c . When the sample is placed in c , then c and c^0 are no longer balanced; a voltage V is induced, where

$$V = \alpha \frac{dM}{dB} \frac{dB}{dt}. \quad (8.34)$$

Here α is a parameter depending on the geometry of the coil and sample. The de Haas–van Alphen oscillations occur in dM/dB and thus will be visible in V .¹⁰

The field is provided in two ways.

- For fields of up to ~ 20 T, superconducting magnets are generally used. The magnet consists of two parts, a large outer coil providing a static field and an inner coil which provides a small modulation field (i.e. dB/dt) of mT amplitude at a few tens of Hz.

⁸See e.g. *Equilibrium thermodynamics* by C.J. Adkins, third edition (Cambridge University Press, Cambridge 1983) page 6 and Appendix.

⁹For enthusiasts, a relatively recent example of this is given in 'A numerical model of quantum oscillations in quasi-two-dimensional organic metals in high magnetic fields', by N. Harrison *et al.*, *Physical Review B* **54**, 9977 (1996).

¹⁰de Haas–van Alphen oscillations are also sometimes observed using a *torque magnetometer*; see J.S. Brooks *et al.*, *Rev. Sci. Instrum.* **64**, 3248 (1987).

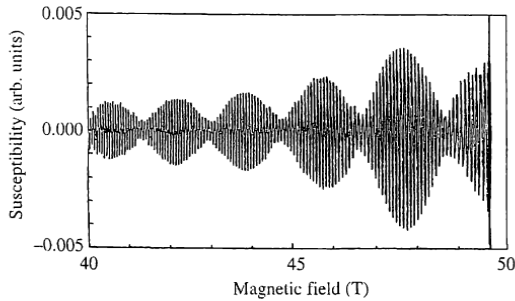


Fig. 8.9 de Haas-van Alphen oscillations in Pt at 4.2 K with $\mathbf{B}$ parallel to [111]. The data have been recorded using a pulsed magnetic field. Note the presence of two frequencies of oscillations due to two extremal orbits about the Fermi surface. The y axis, labelled 'susceptibility', represents the voltage V induced in the coil divided by (dB/dr) , leaving a quantity proportional to the differential susceptibility (dM/dB) (see eqn 8.34). (Data from S. Askenazy, *Physica B* **201**, 26 (1994).)

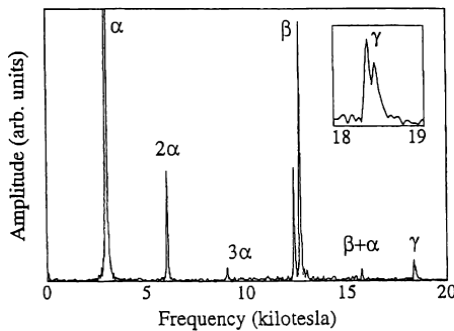


Fig. 8.10 Fourier transform of de Haas-van Alphen oscillations of Sr_2RuO_4 . Fourier transformation is routinely used to help understand de Haas-van Alphen oscillations. The oscillation data (recorded digitally) are processed so that they are equally spaced in $1/B$; the resulting file is Fourier transformed using a desk-top computer. Each extremal cross-section of the Fermi-surface generates a peak in the Fourier transform, the frequency of which is proportional to the k -space area. Oscillations which are non-sinusoidal in shape generate harmonics in the Fourier transform; frequency mixing effects also occur, resulting in sum frequencies. Three Fermi-surface cross-sections (labelled α , β and γ) are shown, along with some harmonics of α and a sum frequency $\beta + \alpha$. (Data from A.P. McKenzie *et al.*, *J. Phys. Soc. Jpn.* **67**, 385 (1998)).

- Pulsed magnets are used for fields of up to ~ 60 T; in this case the magnetic field rises to its maximum value in a few milliseconds, so that dB/dr is large (there is no need for an additional modulation coil).

Some typical de Haas-van Alphen data are shown in Figs. 8.9 and 8.10.

The coils are often mounted on a rotation stage so that the sample can be rotated *in situ*, allowing measurements to be made with $\mathbf{B}$ applied at several orientations to the crystal axes. The coils and sample are mostly mounted in ^3He cryostats or dilution refrigerators, as mK temperatures are needed to observe the oscillations in many cases (see below).

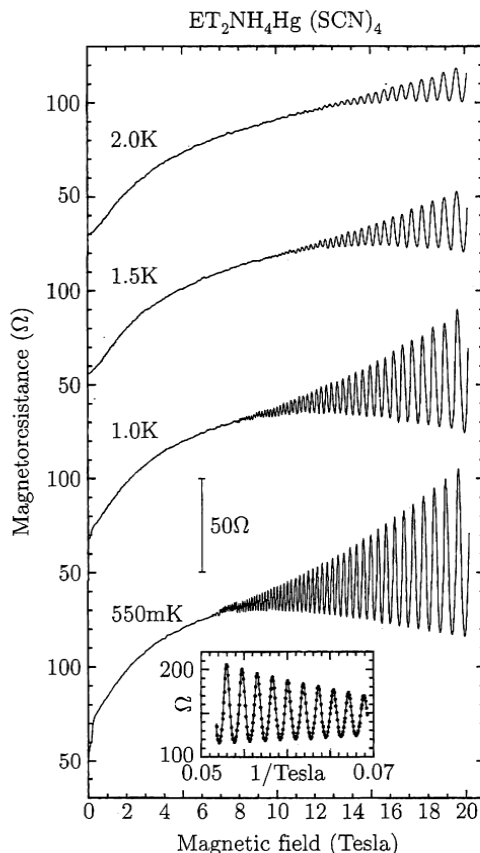


Fig. 8.11 Shubnikov-de Haas oscillations in the organic molecular solid α -(BEDT-TTF) $_2$ NH $_4$ Hg(SCN) $_4$; data for different temperatures have been offset for clarity. The inset shows that the oscillations are periodic in $1/B$. (Data from J. Singleton, *Reports on Progress in Physics* **63**, 1111 (2000).)

8.4.3 Other parameters which can be deduced from quantum oscillations

So far we have concentrated on the information about the Fermi-surface shape provided by the quantum oscillations. However, other valuable information can be deduced. Figure 8.11 shows Shubnikov-de Haas oscillations in an organic solid (see Chapter 7 for more details of this sort of conductor). Two things are immediately obvious.

(1) **The oscillations grow in amplitude as the temperature is lowered.**

This occurs because finite temperatures ‘smear out’ the edge of the Fermi-Dirac distribution function which separates filled and empty states (see Fig. 1.7). At $T = 0$ the Landau tubes will pop out of a very sharp Fermi surface; at finite T the surface will be ‘fuzzy’. The modulation of the density of states around μ caused by the Landau tubes will therefore be less significant. The thermal smearing implies that $\hbar\omega_c$ must be greater than $\sim k_B T$ for oscillations to be observed, i.e. low temperatures are needed.

As the decline in intensity is caused by the Fermi–Dirac distribution function, it can be modelled rather well, and an analytical expression for the temperature dependence of the oscillation amplitude, valid as long as the oscillation amplitude is small, has been derived¹¹

$$\text{Amplitude} \propto \frac{\chi}{\sinh \chi}, \quad (8.35)$$

where $\chi = 14.7 m_{\text{CR}}^* T/B$ (with T in Kelvin and B in Tesla). It is fairly obvious that the parameter χ is proportional to the ratio $k_B T / \hbar \omega_c$; i.e. what we are doing is comparing $k_B T$ and $\hbar \omega_c$. Fits of the oscillation amplitude to eqn 8.35 can therefore be used to give m_{CR}^* .

- (2) **The oscillations grow in amplitude as B increases.** This occurs because scattering causes the Landau levels to have a finite energy width $\sim \hbar/\tau$. As B increases, $\hbar \omega_c$ will increase, so that the broadened Landau levels will become better resolved; theoretical studies have shown that this causes the oscillation amplitude to vary with B as follows (again, the equation is only valid for small oscillations)

$$\text{Amplitude} \propto B^{-r} e^{-\frac{\pi}{\omega_c \tau}}, \quad (8.36)$$

where r is a number ~ 1 . Therefore the scattering rate τ^{-1} can be extracted.

In addition, information about the g -factor of the electrons can be obtained from the oscillations under certain circumstances.¹²

8.4.4 Magnetic breakdown

The frequencies observed in the quantum oscillations correspond to the areas of closed semiclassical orbits about the extremal areas of the Fermi surface. However, under certain circumstances, much higher frequency oscillations can become apparent at high fields, corresponding to larger k -space areas. These are caused by magnetic breakdown.

Figure 8.12 shows a simple Fermi surface cross-section with two open semiclassical orbits (incapable of giving quantum oscillations) and a closed orbit (capable of giving quantum oscillations); the arrows show the electron trajectories. At low fields, one frequency of oscillations corresponding to the closed orbit will be seen. At high fields, the electrons have sufficient cyclotron energy to tunnel in k -space from one part of the Fermi surface to another. Therefore they can now describe much larger closed k -space orbits, leading to higher-frequency quantum oscillations. In effect, the magnetic field is beginning to break down the arrangement of the bandstructure at zero field.

We shall see an example of magnetic breakdown later in this chapter.

8.5 Cyclotron resonance

It is possible to make a direct measurement of ω_c using millimetre-waves or far-infrared radiation to excite transitions between Landau levels. Such an experiment is known as **cyclotron resonance**. The conditions for observing cyclotron resonance in metals and semiconductors are rather different, and the two cases are described below.

¹¹ See *Magnetic oscillations in metals*, by David Shoenberg (Cambridge University Press, Cambridge, 1984) Chapter 2.

¹² See *Magnetic oscillations in metals*, by David Shoenberg (Cambridge University Press, Cambridge, 1984) and J. Singleton, *Reports on progress in physics* **63**, 1111 (2000).

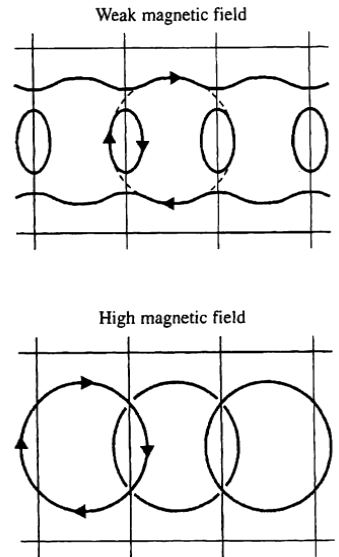


Fig. 8.12 Explanation of magnetic breakdown. Above: semiclassical orbits on a simple Fermi surface at low fields; below: magnetic breakdown at high fields. The magnetic field is perpendicular to the plane of the page.

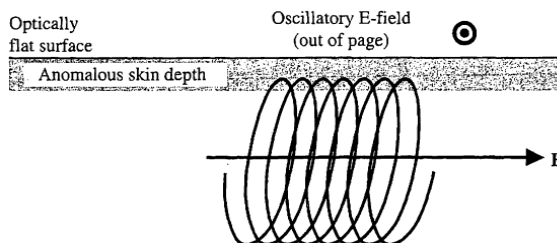


Fig. 8.13 Geometry of a cyclotron resonance experiment in a metal. The shaded region indicates the skin depth for the radiation; the helical orbits of the electrons are shown schematically.

8.5.1 Cyclotron resonance in metals

The field components of electromagnetic radiation decay with distance z into a conducting material as $\exp(-z/\delta)$, where $\delta = (\frac{1}{2}\sigma\omega\mu_r\mu_0)^{-\frac{1}{2}}$ is the *skin depth* or *anomalous skin depth*; here σ is the conductivity of the material, $\mu_r\mu_0$ is its permeability and ω is the angular frequency of the radiation. Metals have rather high conductivities and hence small skin depths; typical values for copper are $\delta \approx 7 \times 10^{-5}$ m at $\omega/2\pi \approx 1$ MHz and $\delta \approx 7 \times 10^{-7}$ m at $\omega/2\pi \approx 10$ GHz.¹³

Typical effective masses in metals combined with the magnetic fields readily available in laboratories have meant that frequencies $\omega/2\pi \sim 1 - 100$ GHz have tended to be applied in cyclotron resonance experiments (see Section 8.8). At such frequencies, radiation cannot penetrate far into the crystal, dictating the geometry of the cyclotron resonance measurement (see Fig. 8.13). The magnetic field is applied parallel to a surface of the crystal, which is placed in a region of oscillating electric field in a resonant cavity; $\mathbf{E}$ of the radiation is arranged to be perpendicular to $\mathbf{B}$ and parallel to the surface. If the frequency of the radiation $\omega = j\omega_c$, where j is an integer, then the electrons will receive a 'kick' from the radiation's electric field every time they come within a skin depth of the surface; this results in absorption of energy. Usually ω is kept constant and the field is swept, so that absorptions are uniformly spaced in $1/B$. As in the case of the quantum oscillations, low temperatures are required. A magnetic field which is very accurately parallel to the sample's surface is required for a successful measurement. This type of experiment is often referred to as the *Azbel-Kaner geometry*.

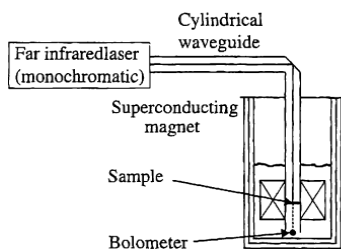


Fig. 8.14 Schematic of a cyclotron resonance experiment in a semiconductor.

8.5.2 Cyclotron resonance in semiconductors

The number densities of carriers in semiconductor samples are much lower than those in metals, so that the radiation can completely penetrate even large samples. A simple transmission arrangement is usually adopted (see Fig. 8.14).

The cyclotron resonance is usually recorded by measuring the sample transmission; experiments are carried out either by fixing the magnetic field and varying the energy of the radiation (see Figure 8.15), or by using a fixed-frequency source (such as a far-infrared laser, which can give a number of monochromatic laser lines in the wavelength range $40 \mu\text{m}$ to $1000 \mu\text{m}$) and sweeping the magnetic field (see Figs. 8.14 and 8.16).¹⁴ The magnetic field is usually provided by a superconducting magnet ($B = 0 - \sim 20$ T). As the whole of the cyclotron orbit experiences the electric field of the radiation (cf. the situation in metals; see above), the quantum-mechanical selection rule

¹³The skin depth is derived in *Electricity and magnetism*, by B.I. Bleaney and B. Bleaney, third edition (Oxford University Press, Oxford, 1989) page 236.

¹⁴An alternative arrangement for semiconductor cyclotron resonance (rather an old-fashioned one) is given in *Electricity and magnetism*, by B.I. Bleaney and B. Bleaney, third edition (Oxford University Press, Oxford, 1989) pages 729-734; the description also provides a more rigorous derivation of the linewidth of the cyclotron resonance than the one given here.

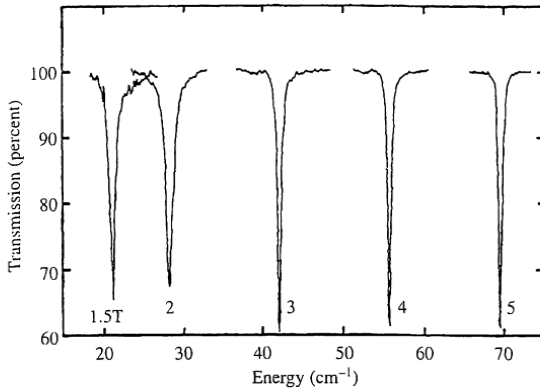


Fig. 8.15 Cyclotron resonances in a GaAs-(Ga,Al)As heterojunction at $T = 4.2$ K; the magnetic field has been applied perpendicular to the two-dimensional electron layer, which has an areal carrier density of $\sim 9 \times 10^{10} \text{ cm}^{-2}$. The data have been recorded by fixing the magnetic field at 1.5, 2, 3, 4, and 5 T; at each field, a Fourier-transform spectrometer has then been used to record the transmission of the sample as a function of energy. (Data recorded by G. Wiggins, University of Oxford. For further data of this type, see e.g. C.J.G.M. Langerak *et al.*, *Phys. Rev. B*, **38**, 13133 (1988).)

for the Landau-level quantum number ($\Delta l = \pm 1$) holds. Thus the resonance condition is

$$\omega = \omega_c = \frac{eB}{m_{\text{CR}}^*}, \quad (8.37)$$

where ω is the angular frequency of the radiation.

The linewidth of the resonance can give information about the scattering rate τ^{-1} . The scattering induces a frequency uncertainty $\Delta\omega_c \sim \tau^{-1}$. If the experiment is a fixed-frequency, swept-field one, this translates to an uncertainty (i.e. resonance width) in magnetic field of $\Delta B = \Delta\omega_c m_{\text{CR}}^* / e \sim m_{\text{CR}}^* / e\tau$.¹⁵

In the case of degenerate semiconductor systems (e.g. heterojunctions), free carriers are present even at low temperatures; typical cyclotron resonance data are shown in Fig. 8.15. However, in lightly doped samples, the carriers must be excited into the bands by either raising the temperature to cause the impurities to ionise (but not so far as to broaden the Landau levels) or by additionally illuminating the sample with above-band-gap radiation.

Figure 8.16 shows an example of the latter technique applied to a single crystal of Ge. In addition to separate resonances caused by light and heavy holes, three electron cyclotron resonances are observed. This occurs because the anisotropic conduction-band minima in Ge lie along the [111] axes (see Fig. 6.5); in Fig. 8.16 the static magnetic field makes three different angles with the four [111] axes. The conduction-band minima in Ge can be described using an effective mass tensor (see Chapter 5); if one of the principal axes of the coordinate system is directed along a [111] axis, then the tensor is diagonal. Under such circumstances, eqn 5.28 shows that the cyclotron resonance occurs at $\omega = \frac{eB}{m^*}$, with

$$m^* = \left(\frac{B^2 m_1 m_2 m_3}{m_1 B_x^2 + m_2 B_y^2 + m_3 B_z^2} \right)^{\frac{1}{2}},$$

where the components of the magnetic field (B_1, B_2, B_3) are defined with respect to the principal axes which make the tensor diagonal. Thus, the cyclotron resonance frequency depends on the orientation of the static magnetic field

¹⁵ Provided that $m_{\text{CR}}^* \approx m^*$, the carrier effective mass for linear motion (and this is often true in direct-gap semiconductors like GaAs) it is apparent that $1/\Delta B \sim$ the carrier mobility (see eqn 9.12).

Fig. 8.16 Absorption caused by cyclotron resonance of electrons and holes in a single crystal of Ge; the electrons and holes are present because the sample has been illuminated by above-band-gap light. The microwave radiation frequency has been fixed at 24 GHz and the magnetic field, applied in a (110) plane at 60° to a [100] axis, has been swept. Cyclotron resonances due to light (low field) and heavy (high field) holes are visible, as are three electron cyclotron resonances. (Data from Dresselhaus *et al.*, *Phys. Rev.* **98**, 368 (1955).)

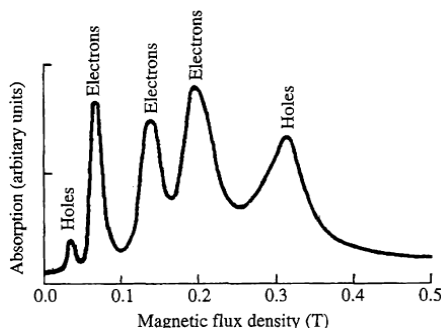


Table 8.1 Effective masses (in units m_e) in Ge and Si determined using cyclotron resonance. The masses m_1 , m_2 and m_3 refer to the components of the effective mass tensors for the conduction band; m_{ICR} and m_{hCR}^* are the cyclotron masses of light and heavy holes respectively.

Material	$m_1 = m_2$	m_3	m_{ICR}	m_{hCR}^*
Ge	0.082	1.64	0.044	0.3
Si	0.19	0.98	0.16	0.3

with respect to the crystal axes. Such angle-dependent cyclotron resonance experiments can be used to determine components of the effective mass tensor.

In the case of Ge and Si, it is found that two of the principal values of the effective mass tensor are equal, so that the constant energy surfaces are ellipsoids with circular cross-sections (see Fig. 6.5). The effective masses of Si and Ge determined by cyclotron resonance are shown in Table 8.1.

8.6 Interband magneto-optics in semiconductors

The application of a magnetic field to a semiconductor splits both the valence and conduction bands into Landau levels, and optical transitions can be induced between hole and electron Landau levels. In reasonably simple systems, the selection rule for the Landau level index is $\Delta l = 0$; more subtle selection rules apply in materials with complex bandstructures.¹⁶ Taking the former, simpler case, the transitions therefore have the energies

$$E(l, B) = E_g + \frac{\hbar e B}{m_{\text{hCR}}^*} \left(l + \frac{1}{2} \right) + \frac{\hbar e B}{m_{\text{eCR}}^*} \left(l + \frac{1}{2} \right), \quad (8.38)$$

where $l = 0, 1, 2, 3, \dots$ and m_{eCR}^* and m_{hCR}^* are the cyclotron masses of the electron and the hole respectively (usually the heavy holes dominate, because of their large density of states). Such a case is dealt with in Exercise 8.1.

¹⁶This is a consequence of the angular momentum carried by the photon. The valence bands of many semiconductors are based on atomic orbitals with total angular momentum quantum number $J = 3/2$, whereas the conduction band is based on orbitals with $J = 1/2$. Thus the one unit of angular momentum carried by the photon is required to promote an electron across the band gap, leaving nothing to spare to change the Landau-level quantum number l . By contrast, in intraband magneto-optical transitions (cyclotron resonance), the photon's angular momentum is used to change l by one ($\Delta l = \pm 1$).

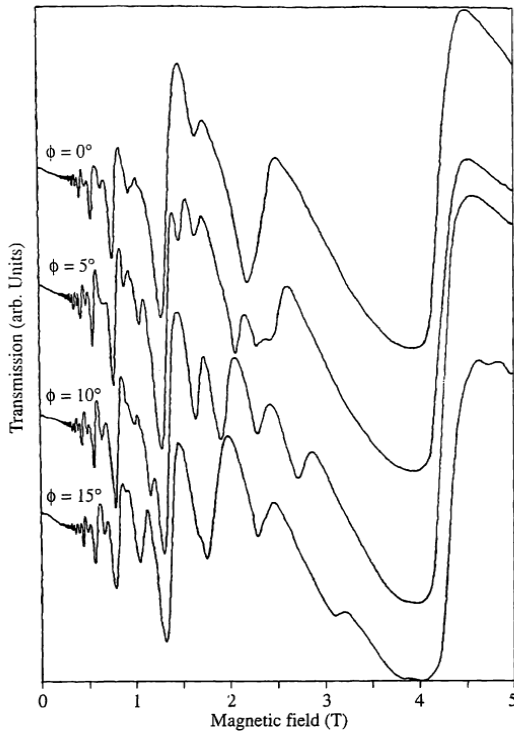


Fig. 8.17 Transmission of PbTe at a fixed photon energy of $h\nu = 0.2077$ eV as a function of magnetic field ($T = 1.9$ K). Data for several orientations of the sample in the external field, which is at all times perpendicular to [111], are shown; ϕ is the angle between the magnetic field and $[\bar{1}10]$. (Data from J. Singleton *et al.*, *J. Phys. C: Solid State Phys.* **19**, 77 (1986).)

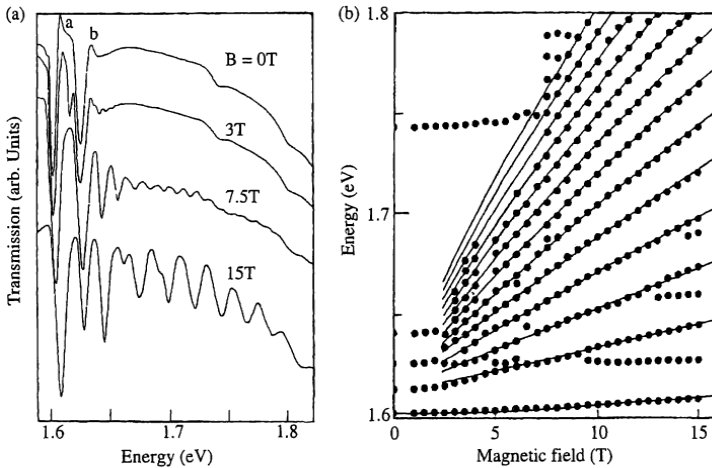


Fig. 8.18 (a) Transmission of a 55 Å GaAlAs quantum well at various fixed magnetic fields ($T = 55$ K). The magnetic field has been applied perpendicular to the plane of the quantum well. (b) Resulting fan diagram of transition energies versus magnetic field; points are data, and the lines are a theoretical fit. (From D.C. Rogers *et al.*, *Phys. Rev. B* **34**, 4002–4009 (1986).)

Illustrative experimental data are shown in Figs. 8.17 and 8.18. In Fig. 8.17, the transmission of a thin film of PbTe has been measured using fixed-frequency radiation of energy $h\nu = 0.2077$ eV as the magnetic field is

swept. Transitions between valence-band Landau levels and conduction-band Landau levels are observed as dips in the transmission; note that the dips get further apart and better resolved as the field increases. Equation 8.38 gives a qualitative understanding of these observations; as the field increases, the inter-Landau-level transitions will successively pass through the radiation energy $h\nu$, resulting in a spectrum which looks roughly periodic in $1/B$. PbTe has complex, anisotropic conduction and valence-band extrema, so that the conduction and valence-band Landau level energies vary as the orientation of the magnetic field changes; this is illustrated by the rapid change in the spectra as the sample is tilted in the field, showing how a comprehensive study of this kind can map out the band shapes.

In systems which involve excitonic effects, there are usually complications to be taken into account. Figure 8.18(a) shows the transmission of a GaAs-(Ga,Al)As quantum well at four fixed fields as a function of energy. At zero field, only the excitonic transitions from the hole subbands to the conduction-band subbands are observed, but as the field increases, transitions between the heavy-hole Landau levels and the conduction-band Landau levels become visible. Figure 8.18 shows a 'fan diagram' of the transition energies versus magnetic field for the same sample. Note the following:

- the Landau-level transitions no longer extrapolate back to a point (cf. eqn 8.38), but instead evolve directly from the various states of the exciton (this is particularly notable for the lowest ($l = 0$ and $l = 1$) transitions); there is a one-to-one correspondence between the hydrogen-like levels of the exciton and the Landau level transitions (1s evolves into $l = 0$ transition, 2s evolves into $l = 1$ transition etc.);
- the field dependences of the transition energies deviate from straight lines at high energies, curving downwards; this is due to the non-parabolicity of the bands as one moves away from the band extrema.

Data such as these can be used to extract the effective masses of the electron and heavy-hole subbands, parameters which describe the nonparabolicity and the exciton binding energy.

A review of recent work in this field is listed under Further Reading.

8.7 Other techniques

The experimental techniques mentioned thus far tend to give information about very restricted regions of the bands concerned. In the case of the magnetic quantum oscillations, electrons very close to the Fermi surface are studied; similarly, the magneto-optical measurements of the semiconductors tend to measure the bands close to their extrema. Furthermore, the methods described above are only usually possible at low temperatures. In the following sections, I shall briefly describe experimental methods which attempt to measure the bandstructure away from these points and which may be used at higher temperatures.

8.7.1 Angle-resolved photoelectron spectroscopy (ARPES)

Angle-resolved photoelectron spectroscopy (ARPES) uses photons with typical energies $h\nu$ in the range ~ 5 eV (ultraviolet) to ~ 1 keV (soft X-ray) to promote electrons from the occupied bands of a solid to unoccupied energy levels from which they can escape. Figure 8.19(a) shows this process schematically. An electron, initially in a state with energy E_i , is excited to an empty state with energy E_f , which is above the vacuum level, E_{vac} ; the electron is then able to escape the solid with kinetic energy

$$E_{\text{kin}} = E_f - E_{\text{vac}} = E_i + h\nu - E_{\text{vac}} \quad (8.39)$$

(this is basically the famous Einstein equation describing the photo-electric effect). By measuring E_{kin} and $h\nu$, one can then deduce E_i .

For many years, angle-integrated photoemission, in which all of the electrons of a certain energy are collected irrespective of angle of emission, was used as a way of characterising the bandstructure of solids. In such an experiment, the number of electrons emitted is basically determined by the density of states $g(E)$ at E_i . However, it was always realised that the wavevector of the emitted electron also contained information about the initial state. With the advent of high-intensity synchrotron radiation sources in the late 1970s and early 1980s, the numbers of emitted electrons became sufficient for angle-resolved collection of electrons to be feasible.

In free space, a single electron cannot absorb a photon, since momentum cannot be conserved. However, in a solid, the extra momentum is provided by the surrounding lattice. It turns out¹⁷ that the component of $\mathbf{k}$ parallel to the surface of the crystal is conserved according to the equation

$$\mathbf{k}_i^{\parallel} = \mathbf{k}_f^{\parallel} + \mathbf{G}^{\parallel}, \quad (8.40)$$

where $\mathbf{k}_i$ is the value of $\mathbf{k}$ for the initial state of energy E_i , $\mathbf{k}_f$ is the final value of $\mathbf{k}$ outside the solid and the suffix $\parallel$ indicates the components parallel to the surface (see Fig. 8.19(b)). Ultraviolet photons have negligible momentum compared to band electrons, so that the photon momentum is neglected in this equation; the lack of periodicity in the direction perpendicular to the surface means that the momentum transfer in this direction is unconstrained. Note that $\mathbf{G}^{\parallel}$ is a *surface* reciprocal lattice vector, that is, one defined by the symmetry of the crystal at the surface. In many cases, the surfaces of crystals are *reconstructed*, and so $\mathbf{G}^{\parallel}$ may be rather different from that of the bulk reciprocal lattice, creating a possible source of error in the interpretation of ARPES. As the energy E_{kin} of the emitted electrons is measured, $\mathbf{k}_f^{\parallel}$ may be deduced using

$$|\mathbf{k}_f^{\parallel}| = \frac{(2m_e E_{\text{kin}})^{1/2}}{\hbar} \sin \theta. \quad (8.41)$$

Equations 8.39, 8.40 and 8.41 are sufficient to reconstruct E_i and $\mathbf{k}_i$ of the initial state.

ARPES has been used extensively with layered compounds in which the bandstructure can be regarded as quasi-two-dimensional; in such materials the deduction of the underlying bandstructure from the emitted electron energies is often relatively simple. Figure 8.20 shows ARPES data from the layered

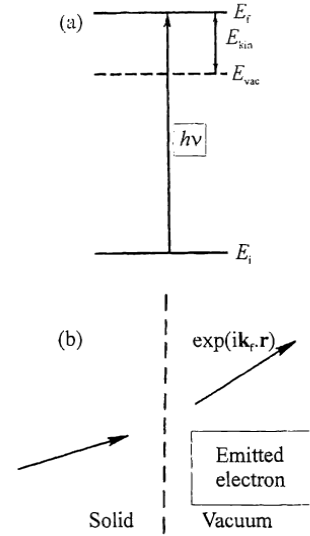


Fig. 8.19 (a) Schematic of ARPES. An electron, initially in a state with energy E_i , is excited by a photon of energy $h\nu$ to an empty state with energy E_f , above the vacuum level, E_{vac} ; the electron is then able to escape the solid with kinetic energy $E_{\text{kin}} = E_f - E_{\text{vac}}$. (b) Simplified geometry of emission process at the surface. The electron is emitted with an energy E_{kin} at an angle θ to the sample surface normal; the component of $\mathbf{k}$ parallel to the surface is conserved according to eqn 8.40.

¹⁷ See F.J. Himpsel, *Advances in Physics* 32, page 11 (1983) for a simple derivation of the conservation laws.

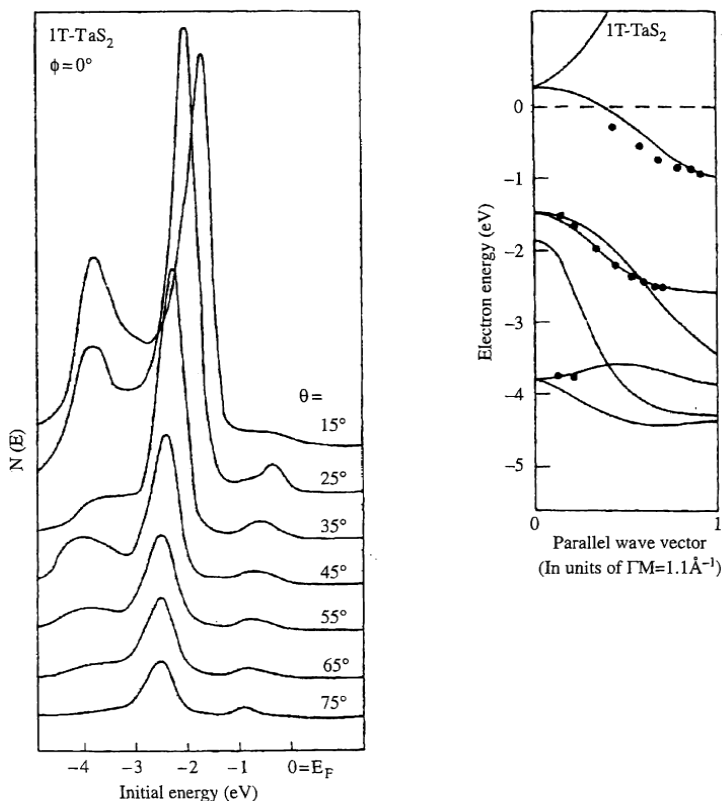


Fig. 8.20 Left: ARPES data from the layered compound 1T-TaS₂; the electron energy has been plotted relative to the Fermi energy E_F . The upper edge of each spectrum gives the position of the Fermi energy. The variation of the peak positions with emission angle maps out the bands of the materials; the values of $(E_i, \mathbf{k}_i)$ deduced (points) are plotted alongside a theoretical bandstructure calculation (curves) on the right-hand side of the figure. (After F.J. Himpsel, *Advances in Physics* 32, 1 (1983).)

compound 1T-TaS₂, which is metallic. Note that the upper edge of each spectrum gives the position of the Fermi energy. The variation of the peak positions with emission angle maps out the bands of the materials; the values of $(E_i, \mathbf{k}_i)$ deduced are plotted alongside a theoretical bandstructure calculation in the right-hand side of the figure.

8.7.2 Electoreflectance spectroscopy

In the 1960s and 1970s electoreflectance spectroscopy (ERS) became a widely used technique for the investigation of the electronic band structure of semiconductors. It is a form of 'modulation spectroscopy' and is the most widely used because of the ease with which electric fields may be applied and modulated in a sample. An experiment involves the measurement of the change in the reflectivity of a sample, caused by changing the electric field at the surface.¹⁸ The field is usually applied with a Schottky barrier arrangement, either by evaporating a very thin metallic layer onto the surface of the semiconductor, or by immersing the sample in an electrolyte (the electrolyte is transparent and being a conductor, behaves in an analogous way to the metal).

¹⁸This is often referred to as the Franz-Keldysh effect. A simple description of the Schottky barrier is given in Section 17.6 of *Electricity and Magnetism*, by B.I. Bleaney and B. Bleaney (OUP, 1989).

When photons interact with a semiconductor, they can cause an electron from an occupied band to be promoted to the conduction band or to a higher unoccupied band. In the case of direct transitions, a 'Joint band diagram' can be drawn in which the energy of the transition is plotted against k within the Brillouin zone. Critical points are defined as (i) minima (ii) saddle points and (iii) maxima in this diagram and are designated M_0 for minima, M_1 and M_2 for saddle points, and M_3 for maxima. From the joint band diagram, a joint density of states can be calculated. Singularities, known as Van Hove singularities, occur in the joint density of states at the energies of the critical points, and appear as abrupt changes in slope.

As an example, consider Fig. 6.6, which illustrates some of the M_0 critical points for GaAs. These are at the energies of (i) the fundamental band-gap of the material, known as E_0 (arrow from band 4 to band 5 at the Γ point), (ii) the gap between the spin-orbit split-off valence band and the conduction band (from band 2 to band 5, also at the Γ point- arrow not shown for clarity), known as $E_0 + \Delta_0$, and (iii) the gaps closer to the L-point known as E_1 and $E_1 + \Delta_1$ (arrows from bands 4 and 3 respectively to band 5).

For a single pair of bands forming an M_0 critical point, no optical transitions can take place below the critical point energy in the absence of an electric field. However, in the presence of an electric field, the conduction and valence bands are 'tipped over' in real space, allowing states from either band to tunnel into the previously forbidden band-gap. Thus there is now a finite chance of an optical transition below the energy of the band-gap, the probability of which falls off exponentially with energy below the band-gap energy. The resulting change in the reflectivity is typically only about 1 part in 10^4 for reasonable modulation fields, and therefore phase sensitive detection techniques are used: an AC field is applied, and a lock-in amplifier only responds to changes in light intensity at the photodetector (which collects the reflected light) at the same frequency as the AC field. The reason why ERS is so attractive for studying electronic structure is that the ERS signal is essentially zero, *except* at the critical points. The energies of the critical points may thus be determined precisely, as well as other useful parameters such as the effective mass. Non-modulation techniques, such as simple reflectivity or absorption measurements give fairly featureless, slowly varying spectra in the region of most critical points, making the determination of their energies impossible from such data.

Figure 8.21 shows ERS data for CdTe, a II-VI semiconductor which possesses a bandstructure very similar to that of GaAs.

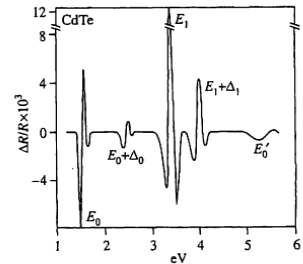


Fig. 8.21 Electroreflectance spectrum of CdTe. Note that the signal is only non-zero at the critical points. (Figure reproduced from M. Cardona, K.L. Shakley and F.H. Pollak, *Phys. Rev.* **154**, 696 (1967))

8.8 Some case studies

8.8.1 Copper

The Fermi surface of copper was determined in a classic series of experiments involving the techniques in this chapter. Enthusiasts might like to look up the original papers, which are listed below.

- A.B. Pippard *Phil. Trans. Roy. Soc.* **A250** 325 (1957) (skin effect measurements leading to the first good model of the Fermi surface).

- D. Shoenberg *Phil. Trans. Roy. Soc.* **A255** 85 (1962) (de Haas–van Alphen measurements); D.J. Roaf *Phil. Trans. Roy. Soc.* **A255** 135 (1962) (theoretical interpretation of de Haas–van Alphen measurements).
- A.F Kip *et al. Phys. Rev.* **124** 359 (1961); J.F. Koch *et al. Phys. Rev.* **133** A240 (1964) (cyclotron resonance).
- B. Stöhr *et al. Phys. Rev. B* **17** 589 (1979) (angle-resolved photoemission and review of more recent work).

8.8.2 Recent controversy: Sr_2RuO_4

Sr_2RuO_4 is an $n = 1$ member of the family of Ruddlesden–Popper oxides of Ru (see Fig. 7.19). It is of great interest, because it has the same crystal structure as $(\text{La}_{1-x}\text{Sr}_x)_2\text{CuO}_4$, an important member of the high- T_c cuprates. In contrast to the latter materials, however, it has been possible to observe magnetic quantum oscillations in Sr_2RuO_4 , and hence make detailed studies of its Fermi surface. The material has a number of other fundamentally interesting properties (e.g. it exhibits an exotic type of superconductivity) and seems likely to continue to be of interest for some time. The material has a relatively simple quasi-two-dimensional Fermi surface, and so the details reported in the papers listed below are followed relatively easily.

In the present context, the studies of Sr_2RuO_4 serve to illustrate the strengths and weaknesses of the various techniques studied in this chapter. ARPES measurements (T. Yokoya *et al.*, *Phys. Rev. Lett.* **76**, 3009 (1996); *Phys. Rev. Lett.* **78**, 2272 (1997); *Phys. Rev. B* **54**, 13311 (1996)) were successful in probing the bands at energies well below E_F and at higher temperatures. However, the Fermi-surface topology implied by the ARPES experiments is clearly at variance with very detailed and unambiguous measurements of de Haas–van Alphen oscillations (A.P. McKenzie *et al.*, *J. Phys. Soc. Jpn.* **67**, 385 (1998), C. Bergemann *et al.*, *Phys. Rev. Lett.* **84**, 2662 (2000).) This discrepancy is almost certainly due to a surface reconstruction (see A. Damascelli *et al.*, *Phys. Rev. Lett.* **85**, 5194 (2000)).

For the enthusiast, general reviews of the properties of Sr_2RuO_4 have been given by Y. Maeno *et al.* in *Nature* **372**, 532 (1994) and *Physica C* **282–287**, 206 (1997).

8.8.3 Studies of the Fermi surface of an organic molecular metal

As the Fermi surfaces of these systems are often very simple (see Section 7.4.3), such experiments are rather easy to understand, and will serve as very visual reminders of the material in Chapter 8.¹⁹

The inset in Fig. 8.22 shows a section through the first Brillouin zone and Fermi surface of $\kappa\text{-(BEDT-TTF)}_2\text{Cu(NCS)}_2$ (see Section 7.4.3 for a discussion as to how this is derived).

The main part of Fig. 8.22 shows the resistance of a sample $\kappa\text{-(BEDT-TTF)}_2\text{Cu(NCS)}_2$ as a function of magnetic field for four different temperatures. Several points are apparent;

¹⁹The experiments described in this section are from N. Harrison *et al.*, *J. Phys.: Condensed Matter* **8**, 5415 (1996) and J. Caulfield *et al.*, *J. Phys.: Condensed Matter* **6**, 2911 (1994).

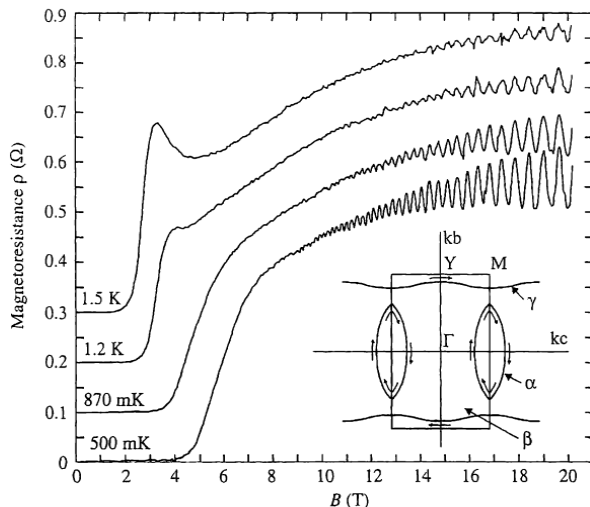


Fig. 8.22 Resistance of κ -(BEDT-TTF) $_2$ Cu(NCS) $_2$ as a function of magnetic field for four different temperatures; the data have been offset for clarity (the resistance is zero at $B = 0$ in all cases). The magnetic field has been applied perpendicular to the highly-conducting planes. The inset shows the Brillouin zone, Fermi surface and the notation α , β and γ used to describe the various Fermi-surface orbits.

- the material is a superconductor (its transition temperature is $T_c \approx 10.4$ K); at low magnetic fields it has vanishing resistance, but sufficiently high magnetic fields drive the substance into its 'normal' resistive state;²⁰
- Shubnikov-de Haas oscillations can be observed in the resistance;
- the amplitude of the Shubnikov-de Haas oscillations declines with increasing temperature.

²⁰See Section H for a brief summary of superconductivity.

The Shubnikov-de Haas oscillations originate from the closed orbits around the hole pocket (labelled α in Fig. 8.22); as described earlier in this chapter, their frequency gives the cross-sectional area of this Fermi-surface section (see eqn 8.32) and the rate at which the oscillations decrease with increasing temperature allows the cyclotron effective mass m_{CR}^* to be estimated (see eqn 8.35). In this case the mass obtained is $m_{\alpha}^* \approx 3.5m_e$.

Figure 8.23 shows that the bandstructure is indeed quasi-two-dimensional in nature; resistance data are shown for four orientations of the sample in the magnetic field. As the angle θ between the magnetic field and the normal to the highly-conducting planes increases, features in the resistance (e.g. the Shubnikov-de Haas oscillations) move to higher total magnetic field as $1/\cos\theta$. This is because the Landau quantisation is chiefly determined by the component of the field perpendicular to the two-dimensional (highly-conducting) planes, i.e. $B \cos\theta$. A particular oscillation will therefore occur at a fixed value of $B \cos\theta$, so that it will move to higher total field as θ increases.

Thus far we have used the data to make deductions about the closed hole pocket, labelled α in the inset of Fig. 8.22. Can anything be deduced about the open sheets along the edges of the Brillouin zone? Fig. 8.24 shows a resistance measurement up to very high magnetic fields made using a pulsed-field magnet. At the highest fields, a higher frequency Shubnikov-de Haas oscillation is visible, superimposed on the oscillations due to the hole pocket. These oscillations are the result of magnetic breakdown between the hole

Fig. 8.23 Resistance of κ -(BEDT-TTF)₂Cu(NCS)₂ as a function of magnetic field at $T = 0.4$ K. Data are shown for four orientations of the sample in the magnetic field; the angle shown is that between the magnetic field and the normal to the highly-conducting planes. The data have been offset for clarity.

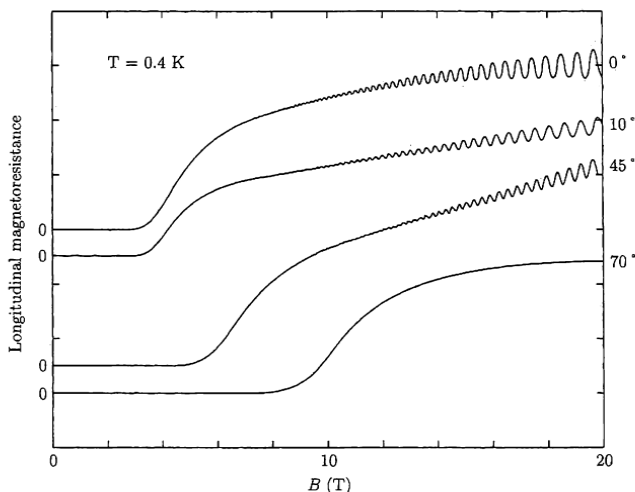
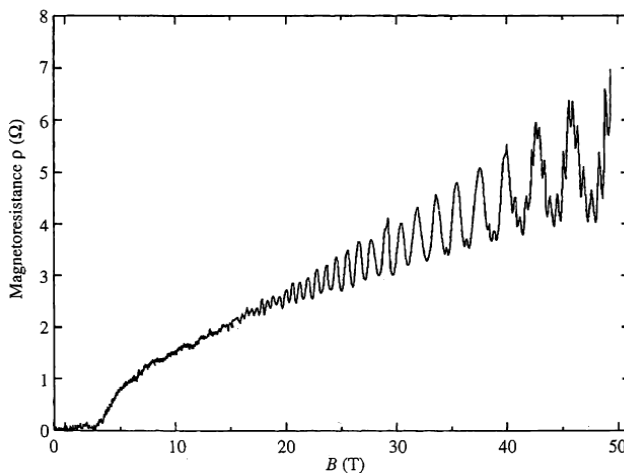


Fig. 8.24 Resistance of κ -(BEDT-TTF)₂Cu(NCS)₂ as a function of magnetic field at $T = 0.5$ K. The data have been measured using a pulsed magnet; the field was applied perpendicular to the highly-conducting planes.



pocket and the open sheets of the Fermi surface (see Fig. 8.12); as a result, the electrons can carry out orbits around the complete Fermi surface (this is known in the trade as the β orbit). The larger k -space area of the breakdown orbit results in the higher frequency of the Shubnikov-de Haas oscillations.

The frequency of the oscillations yields the k -space area of the complete Fermi surface, which (not surprisingly, given the band filling) is equal to the Brillouin-zone area. The temperature dependence can be used to give a cyclotron mass corresponding to the whole Fermi surface, $m_{\beta}^* \approx 6.5m_e$.

We now know an awful lot about the Fermi surface of this material; in the next section, we shall see what happens to the Fermi surface when subtle adjustments are made to the transfer integrals.

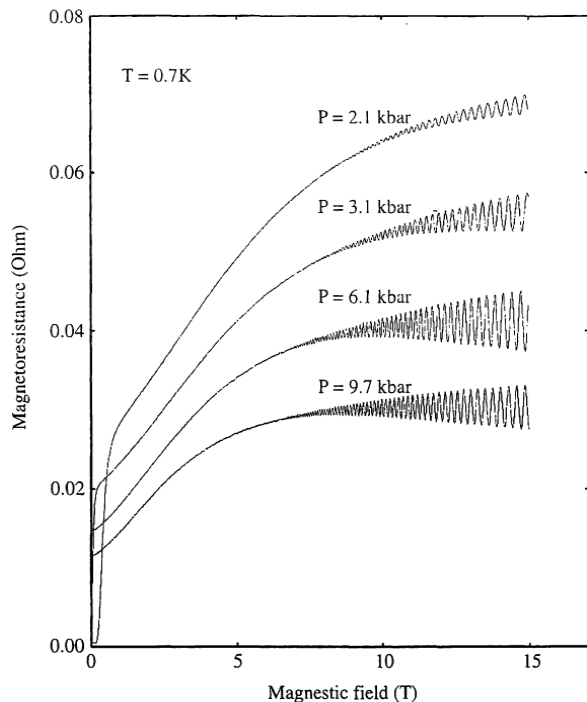


Fig. 8.25 Resistance of κ -(BEDT-TTF) $_2$ Cu(NCS) $_2$ at $T = 0.7$ K for several different hydrostatic pressures.

The effect of hydrostatic pressure

Figure 8.25 shows the resistance of κ -(BEDT-TTF) $_2$ Cu(NCS) $_2$ at $T = 0.7$ K for several different hydrostatic pressures. A comparison with Fig. 8.22 shows that the increasing pressure suppresses the superconductivity; by $P = 3.1$ kbar, the critical field is a fraction of a Tesla. At 6.1 kbar there is no evidence of superconductivity at all. However, the Shubnikov–de Haas oscillations due to the α hole pocket remain present, so that its k -space area and m_α^* can be extracted as a function of pressure. As the pressure increases, the breakdown oscillations become visible at much lower magnetic fields (cf. Fig. 8.26 and Fig. 8.24), allowing the k -space area of the ‘whole’ Fermi surface and m_β^* to be deduced.

Figure 8.27 summarises what happens to the Shubnikov–de Haas oscillation frequencies as a function of pressure. The inset shows that the magnetic breakdown frequency increases by about 6 per cent over the pressure range covered. Section 7.4.3 described how the breakdown orbit corresponds to the complete Fermi surface, which has the same cross-sectional area as the Brillouin zone (as there are two holes per unit cell). The breakdown frequency is therefore directly proportional to the k -space area of the Brillouin zone (see eqn 8.32). As the pressure increases, the real-space unit cell will shrink, and therefore the Brillouin zone will get larger (see eqn 2.7), as observed in Fig. 8.27. The measurement of the magnetic breakdown frequency as a function of pressure is therefore in this case a rather obscure way of deriving the compressibility of the material!

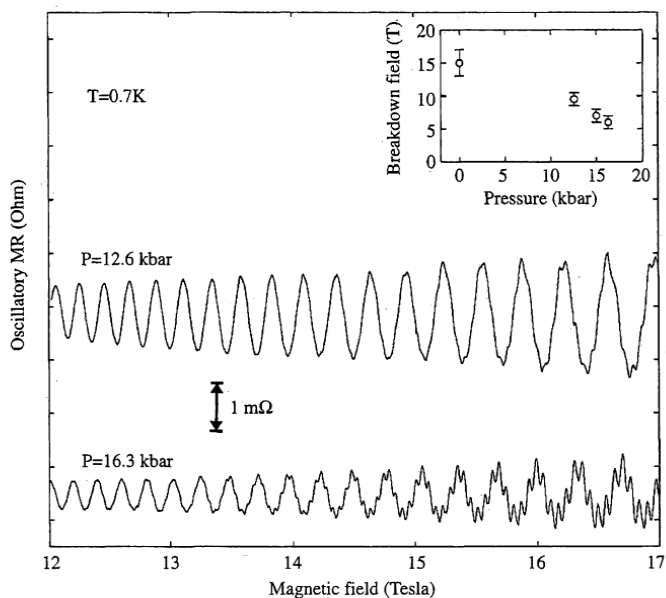


Fig. 8.26 Resistance of κ -(BEDT-TTF)₂Cu(NCS)₂ at $T = 0.7$ K for two large hydrostatic pressures.

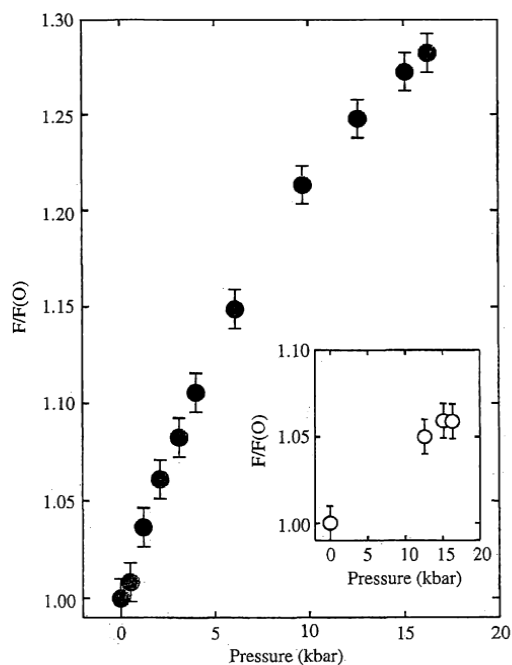


Fig. 8.27 Main figure: the Shubnikov-de Haas oscillation frequency corresponding to the α hole pocket, normalised to its ambient-pressure value, versus pressure. Inset: the pressure dependence of the magnetic breakdown oscillation frequency.

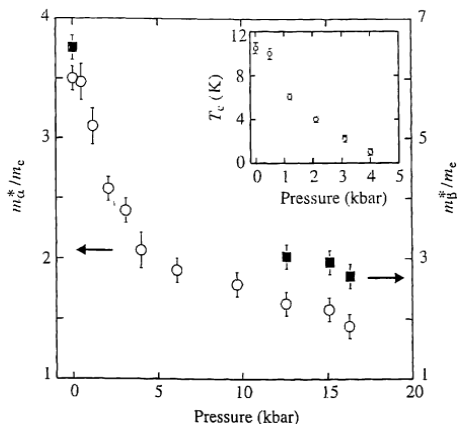


Fig. 8.28 Cyclotron effective masses of the α hole pocket m_{α}^* and 'whole' Fermi surface m_{β}^* as a function of pressure (main figure). The inset shows the superconducting critical temperature T_c as a function of pressure.

The main part of Fig. 8.27 shows that over a corresponding range of pressure, the k -space area (directly proportional to the frequency of the Shubnikov-de Haas oscillations; see eqn 8.32) of the α hole pocket changes by a much greater amount. The pressure is changing the transfer integrals which determine the bandstructure, so that the detailed *shape* of the Fermi surface is gradually altering; another symptom of this is the increasing probability of magnetic breakdown (see Fig. 8.26).

The pressure dependence of the masses

Figure 8.28 shows the pressure dependence of m_{α}^* (corresponding to the α hole pocket), m_{β}^* (corresponding to the 'whole' Fermi surface) and the superconducting critical temperature.

The application of pressure causes both masses to decrease. This is primarily a consequence of the fact that the BEDT-TTF molecules are pushed closer together so that the transfer integrals increase. As shown in Chapter 4, increasing the transfer integrals results in wider bandwidths and smaller effective masses.

The superconducting critical temperature decreases very rapidly as the masses fall; by 5 kbar, T_c is immeasurably small. In Exercise 1.1 we have seen that the density of states at the Fermi energy is proportional to the effective mass; as the effective masses fall with increasing pressure, the density of states at the Fermi energy will also decrease. Mechanisms for superconductivity involve the pairing of electrons of equal and opposite $\mathbf{k}$ caused by the exchange of virtual excitations such as phonons or magnetic fluctuations.²¹ The strength of the pairing is directly determined by the rate at which this exchange can take place, which in turn depends on the density of states at the Fermi energy. Thus, the increase of the transfer integrals causes the effective masses to decrease, which in turn reduces the density of states at the Fermi energy, thereby suppressing the superconductivity.

Experiments such as the one described in this section are being used to test models of superconductivity. Detailed adjustments to the bandstructure are made using an external variable such as pressure; the changed bandstructure is

²¹ See Appendix H and e.g. *Superconductivity* by C.A. Poole, H.A. Farach and R.J. Creswick (Academic Press, San Diego 1995).

then measured using magnetic quantum oscillations. Simultaneous alterations in other properties (e.g. the superconducting T_c) can then be correlated with the bandstructure changes.

8.9 Quasiparticles: interactions between electrons

Measurements of the type described in the previous sections often show that k -space areas of the Fermi-surface sections deduced from the magnetic-quantum oscillation data are in good agreement with the calculated bandstructure. However, if the measured cyclotron effective masses are compared with those predicted from the bandstructure calculations, the agreement is usually much worse.²²

²²Taking an example from the previous section, m_α^* in κ -(BEDT-TTF)₂Cu(NCS)₂ at ambient pressure is $3.5m_e$, whereas the equivalent quantity deduced from bandstructure calculations is $\sim m_e$. In elemental metals, the difference is usually rather smaller; in certain classes of substance, we shall see that the difference can be rather larger.

This is the case in many substances; bandstructure calculations often predict the detailed shape of the Fermi surface rather well but underestimate the observed effective masses. The reason for this can be seen if we consider how we go about a bandstructure calculation; in essence, we constrain ions and molecules to be rigidly fixed in a perfectly periodic arrangement to obtain a periodic potential and hence the bands. However, the ions and/or molecules in a substance will in general be charged, or at the very least possess a dipole moment; as an electron passes through the solid, it will tend to distort the lattice around it owing to the Coulomb interactions between the ions and molecules and its own charge. This leads to the electron being accompanied by a strain field as it moves through the substance. Alternatively one can consider the electron as being surrounded by virtual phonons; the electron and its strain field are often referred to as a 'polaron'. The interactions between the electrons themselves must also be taken into account; as electrons are highly charged, they will repel each other.

By this stage, our initial picture of the electrons as independent particles is beginning to look rather shaky; the electrons interact with each other and carry around a strain field. Landau proposed a way around this problem by supposing that, rather than a system of independent electrons, the solid is populated by a system of independent quasiparticles. He then supposed that the quasiparticles obey Pauli's exclusion principle, so that many of the statistical arguments that we have used to describe the electrical properties of solids still work. But what are the quasiparticles? Landau's argument proposes that as electron-electron interactions are turned on, then the states of the strongly interacting electron system evolve continuously from the states of the independent-electron system.

In the case of an independent-electron system, in the derivations of earlier chapters we have (in effect) specified excited states by counting or listing values of $\mathbf{k}$ ($\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \dots, \mathbf{k}_j$) which denote occupied states with energies above E_F and values of $\mathbf{k}$ that denote empty states with energies below E_F ($\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \dots, \mathbf{q}_l$). The energy of the excited state is then given by adding the energy corresponding to the states described by $\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \dots, \mathbf{k}_j$ to the ground-state energy and subtracting the energies of the states specified by the $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \dots, \mathbf{q}_l$. Now in the case of the independent electron system, we would be able to specify the energies corresponding to the $\mathbf{k}$ and $\mathbf{q}$ easily (for instance in the nearly-free electron model $E(\mathbf{k}) = \hbar^2 \mathbf{k}^2 / 2m_e$).

In the quasiparticle picture, the excited state may still be defined by saying that l quasiparticles have been excited out of states defined by $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \dots, \mathbf{q}_l$ and that j excited quasiparticles have been placed in levels defined by $\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \dots, \mathbf{k}_j$. The energy of the excited state is still given by adding the energy corresponding to the states described by $\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \dots, \mathbf{k}_j$ to the ground-state energy and subtracting the energies of the states specified by the $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \dots, \mathbf{q}_l$. Thus, the excitations of the interacting system behave in a qualitatively similar way to those of the non-interacting system. However, it is now difficult to determine what the E versus $\mathbf{k}$ relationship of the quasiparticles might be.

The theory describing such quasiparticles is often known as Fermi liquid theory; analogously, a non-interacting collection of electrons (as in the Sommerfeld model of Chapter 1) is often referred to as a Fermi gas.

For the present purposes, Fermi liquid theory allows one to treat the effect of the various interactions as renormalising the originally calculated effective mass of the independent band electron to give a (usually heavier) quasiparticle effective mass. The quasiparticles then to all intents and purposes behave as band electrons, but with this modified mass. If m_b is the mass calculated in the bandstructure calculations,²³ then the electron–lattice interactions discussed above result in a mass known as the dynamical mass, m_λ , where

$$m_\lambda \approx (1 + \lambda)m_b, \quad (8.42)$$

where λ is an electron–phonon coupling constant (in many substances, λ is between 0.1 and 1).

Returning to the electron–electron interactions, as electrons are highly charged, they will repel each other. Therefore, as an electron is moved across a sample by an external force, there will be a backflow of electrons caused by this repulsion. Thus, it is ‘harder’ to move electrons about than expected, i.e. the apparent effective mass of the quasiparticles is heavier than m_λ . The effective masses m^* measured in experiments such as the de Haas–van Alphen effect include this contribution from the interactions between the electrons, so that we finally have

$$m^* \approx (1 + F)m_\lambda, \quad (8.43)$$

where F is a constant known as a Fermi liquid parameter.²⁴

Thus, both interactions generally act to increase the mass, thus explaining the discrepancy between bandstructure calculations (which generally predict independent band electrons) and experiments (which measure the mass of the quasiparticles). At a simple level, and in systems such as elemental metals, this need not concern us; we can carry on applying models derived for non-interacting electrons, but dealing instead with excitations called quasiparticles which are a little bit heavier, rather than band electrons; all complications are absorbed in the renormalised effective mass.

However, certain classes of materials display very strong interactions, resulting in extremely heavy quasiparticle effective masses. Compounds containing f-electron metals such as UPt_3 and $\text{Ce}_2\text{Ru}_2\text{Si}_2$ are known as heavy fermion

²³It is important when comparing measured masses with theoretical predictions to compare like with like. For example, the effective masses described by eqn 5.24 are for linear motion, whereas the masses measured in a magnetic quantum oscillation experiment represent orbital averages; the effective mass deduced from a heat capacity experiment is an average over the whole Fermi surface.

²⁴For a detailed treatment of these effects at a reasonably simple level, see Chapters 31–33 of *Many particle theory* by E.K.U. Gross, E. Runge and O. Heinonen (Adam Hilger, Bristol, 1991) or e.g. K.F. Quader *et al.*, *Phys. Rev. B* **36** 156 (1987).

²⁵See, for example, S.R. Julian *et al.*, *Physica B* **199–200**, 63 (1994) and references therein.

systems; effective masses as heavy as 100–1000 m_e have been detected in such substances using magnetic quantum oscillations and other techniques.²⁵

Further reading

A more detailed treatment of the basic ideas involved in magnetic quantum oscillations is given in *Solid state physics*, by N.W. Ashcroft and N.D. Mermin (Holt, Rinehart and Winston, New York, 1976) Chapter 14. Simpler descriptions are in *Electrons in metals and semiconductors*, by R.G. Chambers (Chapman and Hall, London, 1990) Chapter 7, and *Introduction to solid state physics*, by Charles Kittel, seventh edition (Wiley, New York, 1996) Chapters 8 and 9.

For enthusiasts, a splendid and detailed review of magnetic quantum oscillations in metals is found in *Magnetic oscillations in metals*, by David Shoenberg (Cambridge University Press, Cambridge, 1984).

A review of relatively recent cyclotron resonance and interband magneto-optical experiments in semiconductor systems is given in *Optical properties of semiconductors*, Volume 2 of *Handbook on semiconductors*, edited by T.S. Moss and M. Balkanski (Elsevier, Amsterdam 1994), Chapters 6, 7 and 11.

A review of ARPES is given in F.J. Himpsel, *Advances in Physics* **32** 1 (1983); see also M. Grioni *et al.*, *Physica Scripta* **T66**, 172 (1996) and references therein and/or *Fundamentals of semiconductors*, by P. Yu and M. Cardona (Springer, Berlin, 1996), Chapter 8. Further interesting details may be found in articles by E.W. Plummer and W. Eberhardt (*Adv. in Chem. Phys.* **49**, 533 (1982)), and G.J. Lapeyre (*Nucl. Inst. and Meth. in Phys. Research* **A347**, 17 (1994)). The properties of surfaces are reviewed in the book *Electronic properties of surfaces*, edited by M. Prutton (Adam Hilger, Bristol, 1984).

Information on dilution refrigerators and ³He cryostats of the type used for the cryogenic experiments in this chapter is provided in *Experimental low temperature physics*, by A.J. Kent (Macmillan, London 1993), *Matter at low temperatures.*, by P.V.E. McLintock, D.J. Meredith and J.K. Wigmore, (Blackie, Glasgow, 1984) and/or *Practical cryogenics; an introduction to laboratory cryogenics* by N.H. Balshaw (published by Oxford Instruments UK Ltd. and available on request).

Reviews of organic molecular metals are given in *Organic superconductors* by T. Ishiguro, K. Yamaji and G. Saito (Springer-Verlag, Berlin 1998), *Fermi surfaces of low-dimensional organic metals and superconductors* by J. Wosnitza (Springer-Verlag, Berlin 1996) and by J. Singleton in *Reports on Progress in Physics* **63** 1111 (2000).

Interacting electron systems and many-body effects are described in the third edition of the excellent book *Many particle physics*, by Gerald Mahan (Kluwer, Dordrecht, 2000) (highly recommended for enthusiasts) or in *Many particle theory* by E.K.U. Gross, E. Runge and O. Heinonen (Adam Hilger, Bristol, 1991). Two further books which describe classic contributions to this field are *Theory of interacting Fermi systems*, by Philippe Nozieres (Addison Wesley, Reading, 1997), and *Basic notions of condensed matter physics* by P.W. Anderson (Addison Wesley, Reading, 1997).

Heavy fermion systems are reviewed in A.C. Hewson, *The Kondo problem to*

heavy fermions (Cambridge University Press 1993); E.G. Haanappel, *Physica B* **246–247**, 78 (1998) and references therein. The influence of many-body (i.e. electron–electron and electron–phonon) interactions on the de Haas–van Alphen effect has been reviewed by A. Wasserman and M. Springford, *Advances in Physics* **45**, 471 (1996); this review also contains some references to literature on heavy-fermion systems. A concise summary of recent heavy-fermion data is given in Section 8.2 of *Exotic Kondo effects in metals*, by D.L. Cox and A. Zawadowski (Taylor and Francis, London, 1999). Experimental tests for non-Fermi-liquid behaviour in strongly interacting systems are described in V. Vescoli *et al.*, *Science* **281**, 1181 (1998); C. Bourbonnais and D. Jérôme, *ibid* page 1155 and references therein.

Exercises

- (8.1) (a) Figure 8.29 shows $I(9.75)/I(0)$ as a function of photon energy for a sample of a direct-gap semiconductor; the experiment was performed at 77 K. Here $I(B)$ is the intensity of light transmitted by the sample at a magnetic field of B Tesla. Account for the oscillatory behaviour of the plot. Assuming that there are no other strong minima in $I(9.75)/I(0)$ for photon energies below the range shown in the figure, estimate the band gap. Ignore excitonic effects.

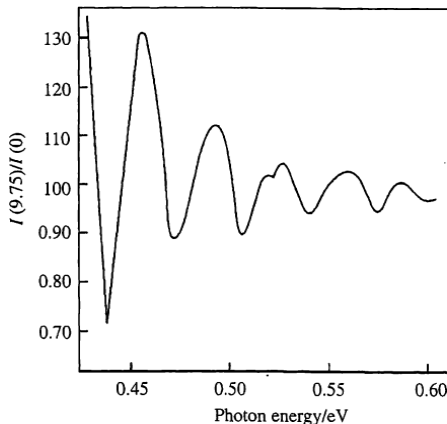


Fig. 8.29 $I(9.75)/I(0)$ as a function of photon energy for a sample of a direct-gap semiconductor; the experiment was performed at 77 K. Here $I(B)$ is the intensity of light transmitted by the InAs sample at a magnetic field of B Tesla.

- (b) A sample of the same semiconductor is lightly doped so as to make it n-type. It is found that the donors have a binding energy of 2.1 meV.

Explain why this energy is almost independent of the donor impurity used and use this fact, along with the data in Fig. 8.29, to estimate the effective masses of the electrons and heavy holes (why *heavy* holes and not *light* holes?). (Assume that the bands are isotropic close to the band extrema, so that $m_{\text{cCR}}^* \approx m_{\text{c}}^*$ and $m_{\text{hCR}}^* \approx m_{\text{hh}}^*$. The relative permittivity of the material is $\epsilon_{\text{r}} = 15.2$.)

- (8.2) Show that in a magnetic field $\mathbf{B}$, the projection of an electron's real-space orbit on a plane perpendicular to $\mathbf{B}$ is its k -space orbit rotated through $\pi/2$ radians about the field direction and scaled by the factor $\hbar e/B$.
- (8.3) Consider the surfaces of constant energy defined by the effective mass tensor of Exercise 5.2. Derive $(\partial A/\partial E)$ for constant energy orbits in planes perpendicular to $\mathbf{e}_3$. Hence show that the cyclotron frequency given by eqns 8.10 and 8.11 for $\mathbf{B}$ parallel to $\mathbf{e}_3$ is the same as that given by eqn 5.28.
- (8.4) Consider the two-dimensional solid of Exercise 3.1. Describe how *electron* and *hole* orbits arise in a magnetic field at certain values of the parameter ν , accompanying your answer with sketches.
- (8.5) A metal is made up of identical atoms arranged in a crystal with the primitive lattice translation vectors $\mathbf{a} = a\mathbf{e}_1$, $\mathbf{b} = ae_2$ and $\mathbf{c} = ce_3$, with $a = 0.3275$ nm and $c = 0.3452$ nm; here $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are the Cartesian unit vectors. The low-temperature magnetisation of the crystal is measured with the magnetic field $\mathbf{B}$ applied parallel to $\mathbf{c}$. Two sets of oscillations periodic in $1/B$ are observed in the magnetisation with periodicities $\Delta(1/B) = 3.526 \times 10^{-5} \text{ T}^{-1}$ and $\Delta(1/B) = 7.590 \times 10^{-4} \text{ T}^{-1}$. However, when the magnetic field is applied parallel to $\mathbf{a}$ or $\mathbf{b}$, no oscillations are seen. Account for these results, and use

them to deduce the dimensions and approximate shape of the Fermi surface. Illustrate your answer with a sketch. Is the metal monovalent or divalent?

- (8.6) A very pure sample of a metal at a temperature of 1 K is illuminated by microwaves of frequency 72 GHz. At this temperature and frequency the skin depth is less than the electron mean free path. A magnetic flux density $\mathbf{B}$ is applied parallel to the surface of the sample and at right angles to the electric vector of the radiation. As B is varied the power absorbed by the sample passes through maxima and minima. Maxima are observed when $nB = 0.95$ Tesla, where n is an integer. Account for this behaviour and deduce the electronic dynamical (cyclotron) mass. Estimate the maximum possible error in the alignment of $\mathbf{B}$ and the sample surface for such an effect to be observed in a typical metal.
- (8.7) A nominally undoped piece of semiconductor is illuminated with above-band-gap radiation and a cyclotron absorption measurement is performed at a frequency of 300 GHz and a temperature of 1.5 K. Cyclotron resonances are observed at 1.1 T and 5.8 T; the width (full width at half maximum) of the lower field resonance is 0.2 T and that of the higher field resonance is 1 T. Estimate the masses and mobilities of the carriers induced by the illumination.

How might the measurement be modified to determine which resonance was due to holes and which was due to electrons?

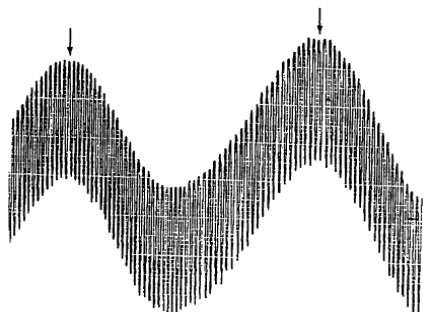


Fig. 8.30 de Haas-van Alphen oscillations in silver.

direction. Silver has a very similar Fermi surface to that of copper. Explain why two frequencies are seen, and use the data to work out the ratio of the radius of the Fermi-surface neck to the radius of the belly.

- (8.9)* A particular monovalent metal has the following properties:
- (a) its Fermi surface is approximately spherical;
 - (b) the Fermi wavevector is $9.1 \times 10^9 \text{ m}^{-1}$;
 - (c) the electrons at the Fermi surface have an effective mass $m^* = 1.3m_e$, where m_e is the mass of a free electron.

Giving brief experimental details, explain how you would deduce properties (a), (b) and (c) using the de Haas-van Alphen effect. Calculate the frequency of the de Haas-van Alphen oscillations that would be observed, the number density of electrons in the metal and its Fermi energy.

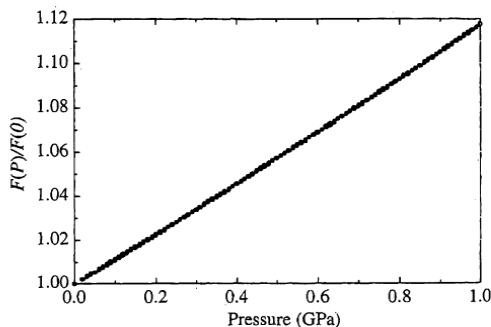


Fig. 8.31 The figure shows $F(P)/F(0)$, the ratio of the de Haas-van Alphen frequency measured with a metal subjected to pressure P to that measured with the metal *in vacuo*.

Figure 8.31 shows $F(P)/F(0)$, the ratio of the de Haas-van Alphen frequency measured with the metal subjected to pressure P to that measured with the metal *in vacuo*. Account for the form of the data, and use the figure to estimate the actual bulk modulus B_{act} of the metal. Calculate the bulk modulus that you would expect for the metal using the Sommerfeld model and comment on your answer.

Describe qualitatively how m^* might be expected to vary with pressure.

- (8.8) Figure 8.30 shows de Haas-van Alphen oscillations in silver with the magnetic field applied along the $[111]$